

Journal Pre-proof

From Carbonate Chemistry to Community Responses: Thematic Evolution in Ocean Acidification and Microbial Research— A Bibliometric Analysis

Yumeng Sun, Yansong Liu, Qing Su, Chengyong Li, Shuheng Zhao, Jingya Zhou



PII: S0141-1136(26)00323-5

DOI: <https://doi.org/10.1016/j.marenvres.2026.108154>

Reference: MERE 108154

To appear in: *Marine Environmental Research*

Received Date: 30 March 2026

Revised Date: 20 May 2026

Accepted Date: 23 May 2026

Please cite this article as: Sun, Y., Liu, Y., Su, Q., Li, C., Zhao, S., Zhou, J., From Carbonate Chemistry to Community Responses: Thematic Evolution in Ocean Acidification and Microbial Research— A Bibliometric Analysis, *Marine Environmental Research*, <https://doi.org/10.1016/j.marenvres.2026.108154>.

This is a PDF of an article that has undergone enhancements after acceptance, such as the addition of a cover page and metadata, and formatting for readability. This version will undergo additional copyediting, typesetting and review before it is published in its final form. As such, this version is no longer the Accepted Manuscript, but it is not yet the definitive Version of Record; we are providing this early version to give early visibility of the article. Please note that Elsevier's sharing policy for the Published Journal Article applies to this version, see: <https://www.elsevier.com/about/policies-and-standards/sharing#4-published-journal-article>. Please also note that, during the production process, errors may be discovered which could affect the content, and all legal disclaimers that apply to the journal pertain.

© 2026 Published by Elsevier Ltd.

From Carbonate Chemistry to Community Responses: Thematic Evolution in Ocean Acidification and Microbial Research— A Bibliometric Analysis

Yumeng Sun¹, Yansong Liu¹, Qing Su², Chengyong Li¹, Shuheng Zhao¹ and Jingya Zhou^{1,*}

1 College of Geography and Ocean Sciences, Yanbian University, Hunchun 133300, China

2 College of Engineering, Yanbian University, Yanji, 133002, China

*Correspondence: 0000008750@ybu.edu.cn

Abstract: Ocean acidification (OA) is reshaping marine biogeochemistry and threatens microbial communities that regulate carbon and nutrient cycling. Although existing bibliometric reviews have examined OA in relation to coral reefs, calcifying organisms, and broader marine ecosystems, no study has systematically mapped the specific sub-domain of OA impacts on microbial ecology, a gap that hinders identification of methodological blind spots, collaboration imbalances, and under-explored research frontiers unique to microbial systems. Meanwhile, research progress on OA-driven microbial change remains fragmented and lacks a systematic analysis of the field's evolutionary trajectory and emerging frontiers. This study presents a comprehensive bibliometric analysis of 495 publications retrieved from the Web of Science Core Collection (2005–2025), utilizing CiteSpace to map the knowledge domain of OA impacts on microbial ecology. Temporal analysis reveals three distinct developmental phases: emergence (2005–2010), exponential growth (2011–2021), and recent stabilization (2022–2025). The global collaboration network spans 53 countries, characterized by a triadic leadership structure involving China, the United States, and Germany, with the GEOMAR Helmholtz Centre serving as a central institutional hub. Keyword co-occurrence and burst detection analyses uncover a significant paradigm shift: the research focus has transitioned from foundational carbonate chemistry parameters to complex ecosystem-relevant microbial processes, including community structure, functional genes, and biogeochemical cycling. Notably, “responses” emerges as the most active contemporary research frontier with the strongest recent citation burst, reflecting a consolidated focus on how microbial communities adapt to acidification stress at physiological, community-structural, and functional levels. However, network analysis also reveals structural blind spots: archaea and viral ecology remain conspicuously absent from high-frequency keyword clusters despite their recognized ecological importance, and research contributions from Africa, Southeast Asia, and Small Island Developing States are markedly limited. Based on these findings, we propose four evidence-linked strategic directions centered on multi-omics integration, spatiotemporal expansion through global observatory networks, factorial multi-stressor experimental designs, and bridging molecular processes to ecosystem-scale biogeochemical cycles. This study provides a data-driven roadmap for next-generation research on OA-microbe interactions, essential for predicting ecosystem resilience in a changing ocean.

Keywords: Ocean acidification; Microbial ecology; Response; Bibliometric analysis;

38 CiteSpace

39 1. Introduction

40 Ocean acidification (OA), often termed “the other CO₂ problem,” is among the most pressing
41 environmental challenges of the twenty-first century (Doney et al., 2020; Schönberg et al.,
42 2017). Anthropogenic CO₂ emissions have measurably decreased ocean pH(Doney, Fabry, et
43 al., 2009a; Feely et al., 2004); since the onset of the Industrial Revolution, mean oceanic pH
44 has declined by ~0.1 units(Martins Medeiros & Souza, 2023). As the ocean absorbs ~25-30%
45 of atmospheric CO₂, carbonic acid forms and marine carbonate chemistry is altered. Under
46 current emission trajectories, ocean pH is projected to decline by a further 0.3-0.4 units by 2100
47 (Jiang et al., 2023). This unprecedented rate of chemical change, occurring at a pace roughly
48 100 times faster than background variability over the past ~20 million years, poses profound
49 challenges for organisms and ecosystems that have evolved under relatively stable chemical
50 conditions(Fabry et al., 2008).

51 Marine microorganisms—including bacteria, archaea, protists, and phytoplankton—form the
52 foundation of oceanic food webs and drive biogeochemical cycles that regulate climate and
53 nutrient distributions(Glibert & Mitra, 2022; Levine et al., 2025). These taxa account for
54 roughly 50% of global primary production and are integral to carbon, nitrogen, phosphorus,
55 and sulfur cycling(Abreu et al., 2022; E. L. Madsen, 2011). Consequently, the structural and
56 functional responses of microbial communities to changing carbonate chemistry have far-
57 reaching implications for ecosystem stability, fisheries productivity, and global carbon-cycle
58 dynamics(Azam & Malfatti, 2007; Wang et al., 2024). As evidence accumulates for the
59 sensitivity of microbial processes to carbonate-system perturbations, understanding OA
60 impacts on microbial ecology has become increasingly urgent(Marín-Samper et al., 2024;
61 O'Brien et al., 2016).

62 Deciphering how OA shapes microbial life requires confronting a multi-scale, multi-stressor
63 problem. OA-microbe interactions are modulated by chemical, biological, and ecological
64 processes that span spatial and temporal scales(Gao et al., 2020; Liu et al., 2010). Perturbations
65 in pH and carbonate chemistry can directly affect microbial physiology, influencing processes
66 such as calcification in coccolithophores, nitrogen fixation in diazotrophs, and taxon-specific
67 metabolic rates(Meron et al., 2012; Piontek et al., 2010; Riebesell et al., 2000). At the cellular
68 level, shifts in pH and carbonate-ion availability affect membrane stability, enzyme kinetics,
69 and metabolic pathways, while altered CO₂ levels modulate carbon-fixation strategies and
70 energy allocation. Community-level responses include changes in species composition,
71 competitive interactions, and functional diversity that cascade through food webs and reshape
72 ecosystem services (Brussaard et al., 2013; Kuffner et al., 2007; Nelson et al., 2020).
73 Importantly, these responses frequently co-occur with warming, nutrient-regime shifts, and
74 deoxygenation, creating multiple-stressor contexts that challenge single-factor inference and

complicate cross-study comparison(Griffen et al., 2016; Gunderson et al., 2016; Riebesell & Gattuso, 2014). Furthermore, OA can alter virus–host interactions in marine ecosystems, with evidence from naturally acidified Mediterranean benthic sites showing that viral life strategies shift from predominantly lytic to temperate (e.g., chronic infection) under low-pH conditions, thereby reducing viral-induced prokaryotic mortality and reshaping organic matter cycling (Tangherlini et al., 2021).

Over the past two decades, OA–microbe research has expanded rapidly across laboratory experiments, mesocosm studies, field observations, and modelling approaches(Hu et al., 2021; Ridgwell et al., 2009; Wang et al., 2023). Short-term microcosm experiments have demonstrated that even modest pH reductions can significantly modify the structure and diversity of bacterial communities, including key taxa involved in nitrogen cycling in oxygen-minimum zones(P. Aguayo et al., 2020).Investigations have addressed effects on individual microbial taxa, community-level responses, and ecosystem-scale consequences across diverse marine environments, from polar systems to tropical coral reefs(Wang et al., 2023; Webster et al., 2016; Witt et al., 2011). This rapid growth has generated a large and heterogeneous evidence base, but also a practical bottleneck: results are distributed across disciplines, organisms, habitats, endpoints, and experimental designs, making it increasingly difficult to (i) identify the intellectual foundations that structure the field, (ii) pinpoint where evidence is concentrated versus where mechanistic or geographic gaps persist, and (iii) detect emerging priorities driven by the urgent need to predict microbial-mediated biogeochemical feedbacks under OA and co-occurring stressors. Given the complexity of microbial responses to OA and the intrinsically interdisciplinary nature of this topic, a comprehensive, field-specific synthesis is warranted(X. Zhang et al., 2025).

Despite the centrality of microorganisms to OA impacts at ecosystem scale, a targeted bibliometric analysis focused specifically on OA effects on microbial ecology is lacking. Existing bibliometric reviews have examined OA in relation to coral reef ecosystems(Y. Q. Geng et al., 2024) , calcifying organisms, and broader marine biodiversity patterns (N. Mao et al., 2023). However, these studies do not isolate the microbial sub-domain, which differs fundamentally in (i) the taxonomic groups investigated (bacteria, archaea, protists, cyanobacteria vs. macro-organisms), (ii) the experimental approaches employed (mesocosm CO₂ enrichment, metagenomics, single-cell techniques), and (iii) the biogeochemical processes emphasized (nitrogen fixation, carbon concentrating mechanisms, dissolved organic matter cycling). This absence of a microbiology-specific mapping limits a quantitative understanding of who collaborates with whom, which institutions serve as hubs, how concepts have shifted from carbonate chemistry descriptors toward ecological and functional mechanisms, and which topics are rising most rapidly. Without such a field-specific mapping, it is harder to coordinate cross-institutional efforts, avoid duplicated lines of inquiry, and transparently prioritize research

directions that can best reduce uncertainty in ecosystem forecasts and monitoring strategies.

Conventional narrative reviews, while valuable for in-depth mechanistic discussion, cannot systematically quantify collaboration structures, detect cross-disciplinary blind spots, or track the temporal trajectory of concept emergence across hundreds of publications. A bibliometric approach is therefore required to answer the structural and evolutionary questions posed above. The present study addresses this knowledge gap by conducting a comprehensive bibliometric analysis of global research on OA impacts on microbial ecology, spanning the period from 2005 to 2025. This study addresses two critical research questions through a systematic bibliometric investigation:

RQ1: How has the OA–microbial ecology literature evolved from 2005 to 2025 in terms of publication dynamics, collaboration patterns, and intellectual bases — and do these structural features reveal discipline-specific collaboration gaps or geographic imbalances that constrain the field’s progress?

RQ2: What thematic structures and emerging frontiers can be identified from co-citation/keyword networks and burst detection, and to what extent does the bibliometric structure reveal methodological or taxonomic blind spots that warrant targeted future investigation?

By linking these quantitative patterns to the scientific challenges of multi-scale, multi-stressor OA–microbe responses, this study provides an evidence-based synthesis of where progress is robust, where knowledge gaps remain, and which directions are most strategic for next-generation research.

2. Data and Methodology

2.1 Data Source and Retrieval Strategy

The bibliometric analysis was conducted using the Web of Science (WoS) Core Collection database, selected for its comprehensive coverage of high-quality peer-reviewed literature and standardized bibliographic information essential for advanced bibliometric analyses (Jia et al., 2023). The Web of Science database provides consistent metadata across multiple disciplines and time periods, ensuring reliable identification of citation relationships, author affiliations, and journal classifications necessary for network construction and temporal evolution tracking (Pranckutė, 2021).

The literature search strategy was designed to capture research publications specifically addressing the intersection of OA and microbial ecology, while maintaining sufficient specificity to exclude tangentially related studies. Since WoS records show that the first manuscript involving the “Effect of Ocean Acidification on Microbial Ecology” was published in 2005 (Orr et al., 2005), we limited the scope of the study from 2005 to 2025. To ensure

retrieval completeness and precision, the search strategy incorporated controlled vocabulary and natural language terms related to “ocean acidification” and “microbial ecology”, with the Boolean search query being iteratively refined through scope validation protocols as follows:

TI=(“ocean acidification” OR “seawater acidification”) AND TS=(“microb*” OR “plankton” OR “phytoplankton” OR “bacterioplankton” OR “coccolithophore*” OR “cyanobacteria” OR “diatom*” OR “microbial community”)

Duplicates were removed via CiteSpace’s built-in deduplicator. Three independent reviewers assessed title/abstract relevance using the inclusion/exclusion criteria: (1) Document types: Articles, reviews, conference proceedings. (2) Language: English-only. (3) Topical relevance: Studies must address both (a) ocean acidification or seawater acidification as a primary research focus AND (b) impacts on or responses of marine microbial organisms, communities, or processes. (4) Exclusions: Totally irrelevant fields such as freshwater acidification, terrestrial soil chemistry, biomedical engineering, and studies focusing exclusively on macro-organisms without microbial components. After manual screening and deduplication, we ultimately obtained 495 articles related to OA impacts on microbial ecology as valid articles, which were then exported in the “RefWorks” plain text format with full metadata. The exported literature covers 20 years and records the emergence, expansion and maturity stages of marine acidification studies.

2.2 Bibliometric Analysis Framework

Bibliometric analysis provides a rigorous framework for characterizing the evolution, structure, and trends of scientific domains by examining publication trajectories, citation networks, co-authorship, and keyword co-occurrence(Chen, 2005).CiteSpace, a widely used visualization platform, enables mapping of knowledge domains and detection of emerging topics through network and temporal analyses(Chen, 2017) . Applied to OA–microbe research, these tools can delineate major knowledge clusters, identify influential works and collaboration hubs, and reveal burst topics that signal shifting priorities in marine microbial science (Y. Geng et al., 2024; Ning Mao et al., 2023). In this study, we employed CiteSpace version 6.3.R3 Advanced, selected for its ability to integrate multiple analytical approaches — including co-citation mapping, keyword clustering, collaboration network construction, and citation burst detection — within a unified framework that enables comprehensive examination of research landscapes from complementary perspectives(Chen, 2004).

The analysis proceeded through four primary modules: (1) Co-citation Network Analysis: Mapping relationships among frequently co-cited references, authors, and journals to reveal the intellectual backbone of the field; (2) Keyword Co-occurrence and Clustering: Identifying dominant research themes, their interconnections, and temporal evolution; (3) Collaboration Network Analysis: Examining the geographic, institutional, and author-level structure of

scientific cooperation. (4) Citation Burst Detection: Detecting terms and references that have experienced sudden increases in attention, signaling research frontiers or major thematic transitions.

The analytical process was carried out in several sequential steps. First, a new CiteSpace project was created, and the complete records and cited reference data retrieved from the Web of Science Core Collection were imported into the workspace. Second, the relevant parameters were configured, including setting the time slicing to one-year intervals, with results from individual years subsequently merged to form an integrated view. Node types for analysis—such as author, keyword, journal, category, institution, country, and cited reference—were selected according to the specific research questions. For keyword and author visualizations, functions such as co-occurrence mapping, clustering, and burst detection were applied to reveal dominant thematic areas, track their evolution over time, and detect research hotspots. Likewise, author, institution, and country co-occurrence networks were constructed to map patterns of scholarly collaboration, with thresholds and pruning methods adjusted to highlight the most influential actors and partnerships.

Finally, through incorporating CiteSpace's Modularity (Q-value) and Silhouette (S-value) indicators, along with insights from citation bursts and cluster label interpretations, we conduct a detailed examination of node magnitude, linkage density, and key standout features in the visualization. This approach not only clarifies connections and intersections across various research themes but also uncovers emergent research hotspots and future directions through highly cited keywords or publications (Wei et al., 2020). By comparing these visual representations both longitudinally and transversely, this study examines the status and evolutionary trajectory of OA and microbial ecology research from the angles of cooperative networks, disciplinary scope, journal distribution, and keyword evolution, thereby offering valuable support for further studies or theoretical expansions.

2.3 Quality Assessment and Validation

Multiple quality assessment measures were implemented to ensure the reliability and validity of bibliometric results. Network modularity (Q-values) and mean silhouette scores (S-values) were calculated for all clustering analyses to assess the coherence and distinctiveness of identified research clusters. Modularity values above 0.3 indicate meaningful community structure, while silhouette scores above 0.7 suggest high cluster quality and internal consistency.

The temporal stability of identified patterns was assessed through parameter sensitivity analysis. Specifically, we varied three key parameters: (i) the top N selection threshold ($N = 30, 50, 70$ per time slice), (ii) the pruning method (Pathfinder vs. MST only vs. no pruning), and (iii) the time-slice width (1-year vs. 2-year intervals). Results showed that the ten-cluster keyword structure was stable across all top N values (cluster labels and membership changed by <8% of

nodes between $N = 30$ and $N = 70$), and the three major author clusters remained intact under all pruning methods. Two-year time slices merged some minor temporal burst signals but preserved all major burst keywords (strength >3.0). Manual verification of the top 20 high-centrality nodes was conducted by tracing original publications to confirm that automated procedures accurately identified key research contributions and collaborative relationships; no misattributions were found.

3. Results and Discussion

3.1. Trends in publications

The temporal distribution of publications in OA and microbial ecology research exhibits a distinctive evolutionary pattern characterized by exponential growth, maturation, and recent stabilization phases spanning two decades (Fig.1). The initial emergence phase (2005-2010) demonstrates the field's nascent stage, with publication numbers remaining consistently low, ranging from 1-15 papers annually and a cumulative total reaching only 56 papers by 2011. This period represents the foundational years when OA research was establishing its conceptual framework and methodological approaches, with limited but steady scholarly interest reflected in the gradual accumulation of research contributions (Doney, Fabry, et al., 2009b). For example, Breitbarth et al. observed through the Mid-Universe experiment that surface seawater acidification led to increased iron bioavailability (Breitbarth et al., 2010). The modest publication volume during these early years aligns with the typical pattern of emerging scientific disciplines, where initial investigations focus on establishing basic principles and experimental protocols rather than generating large volumes of empirical studies. At this stage, the potential of microorganisms to adapt to OA, whether through genetic changes at the species level or species succession at the community level, is unknown. Therefore, it is necessary to elucidate the effects of OA on key species and microbial diversity (Liu et al., 2010).

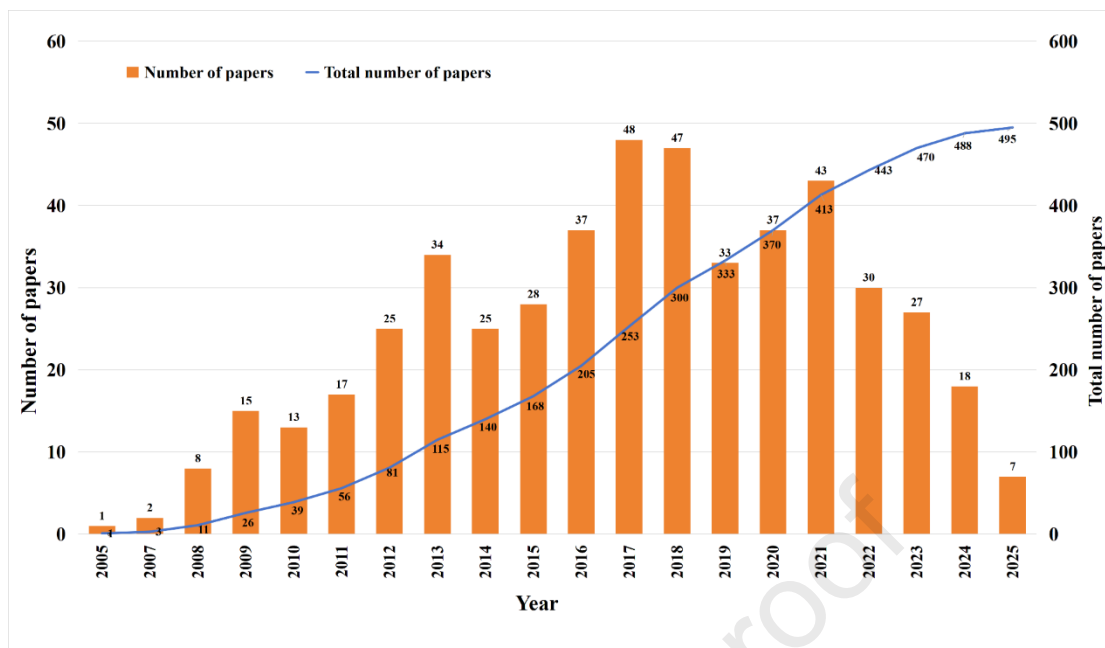


Figure 1. Number of articles published per year from 2005 to 2025.

The rapid growth and fluctuation phase (2011-2021) can be further divided into two sub-periods. From 2011 to 2017, annual output increased substantially from 17 to 48 papers, reflecting the field's expansion from a specialized niche to a recognized research area within marine science. However, from 2018 to 2021, output fluctuated between 33 and 47 papers per year without sustained growth, suggesting that the field's initial expansion momentum had begun to plateau before the more pronounced decline observed after 2021. At this time, this phase of research has been extensively studied in various oceans around the world, with Hancock et al. showing that OA has a negative impact on autotrophs in the Southern Ocean (Hancock et al., 2020), while Wang's research in the Arctic Ocean showed that OA significantly changed the functional gene structure and evenness of the Arctic Ocean bacterioplankton community (Wang et al., 2023). The results of an in-situ short-term microscopic experiment in the South China Sea show that plankton communities in this subtropical coastal waters are insensitive to OA in terms of primary production and nutrient transfer from phytoplankton to zooplankton (Wang et al., 2019). The peak production years of 2017-2018 (48 and 47 papers respectively) suggest optimal research momentum, likely driven by increased funding availability, growing environmental awareness, and the establishment of standardized experimental methodologies that enabled more researchers to contribute meaningful investigations. The effects of OA and multiple stressors on microorganisms have also attracted widespread attention, with OA combined with warming and deoxygenation shown to stimulate production in primary producers (Gao et al., 2020).

The recent phase (2022-2025) shows a marked decrease from peak levels, with annual output falling to approximately 18-30 papers per year and the cumulative total approaching 495 by

early 2025. Several non-exclusive factors may account for this pattern. First, research saturation in well-explored sub-topics, such as single-species calcification responses, may have reduced the volume of incremental studies, as prior meta-analyses have already synthesized results from hundreds of individual experiments across multiple taxa (Boyd et al., 2018b; Wittmann & Pörtner, 2013). Second, the field appears to be shifting toward longer-duration, multi-stressor experimental designs that require more time before yielding publications (Boyd et al., 2018b; Gao et al., 2019b); such experiments increasingly demand factorial combinations of temperature, pH, and oxygen, creating logistical complexity that extends project timelines (Riebesell & Gattuso, 2014). Third, disruptions caused by the COVID-19 pandemic (2020–2022) curtailed field campaigns and mesocosm experiments worldwide (Hobday et al., 2024; Köpsel et al., 2021), effects whose publication lag extends into recent years. It should also be noted that the 2024 and 2025 counts are likely provisional, as Web of Science indexing for these years may be incomplete at the time of data retrieval (Larsen & von Ins, 2010; Liu et al., 2024). Therefore, whether the post-2017 trend represents genuine stabilization, a temporary dip, or the early stage of sustained decline cannot be definitively determined from the current dataset alone.

3.2. *The Interdisciplinary Architecture of the Field*

The discipline co-citation network analysis provides crucial insights into the interdisciplinary landscape and knowledge structure underlying OA and microbial ecology research (Fig.2). The visualization reveals a complex network of scientific disciplines that collectively contribute to understanding the multifaceted impacts of OA on marine microbial communities. Notably, the network structure exposes an asymmetry between the field's disciplinary inputs and its methodological integration. "Environmental Sciences" occupies the most prominent position, which is expected given the field's applied orientation. However, the more informative finding is the relative positioning of peripheral nodes: "Limnology" and "Atmospheric Sciences" show weak connectivity to the core cluster despite their mechanistic relevance — limnology for understanding estuarine pH gradients where freshwater–marine microbial transitions occur, and atmospheric sciences for realistic CO₂ forcing scenarios. This pattern is consistent with a structural blind spot in which the field has successfully integrated biological and chemical disciplines but may have underutilized the freshwater–marine continuum perspective and climate modeling inputs. (Salomon & McKechnie, 2025). Whether this reflects genuine intellectual disconnect or simply terminological separation (e.g., estuarine acidification research using different keywords) cannot be determined from co-citation patterns alone and warrants further investigation.

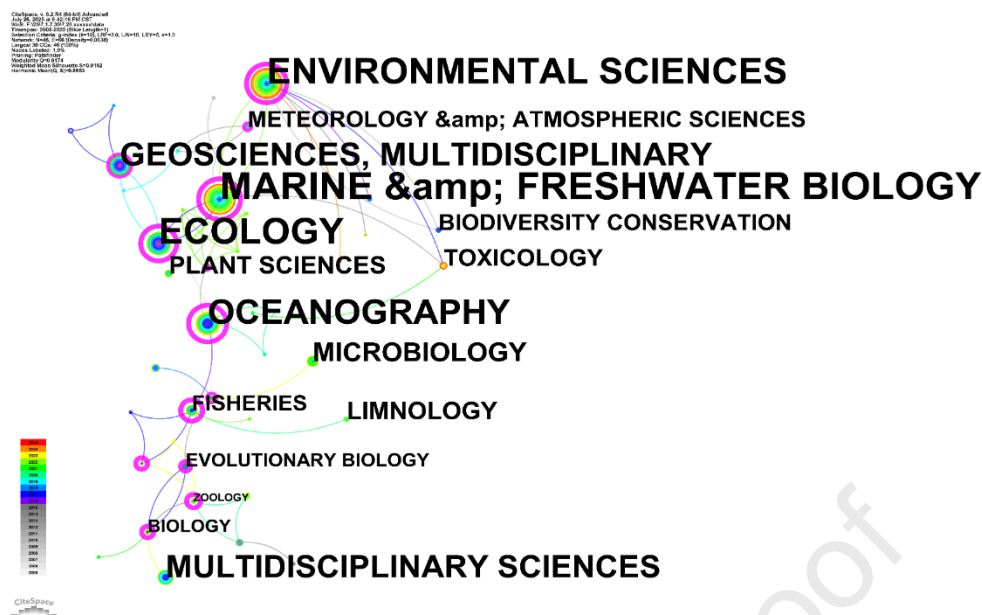


Figure 2. Co-occurrence network diagram for the subject area of acidification-microbial interaction. Node ring colours represent the temporal distribution of publications or citations: cooler colours (blue, green) correspond to earlier years, while warmer colours (yellow, orange, red) indicate more recent activity. Purple rings signify high betweenness centrality.

Oceanography represents another foundational discipline, positioned strategically within the network to bridge physical and biological sciences. The prominence of oceanography reflects the fundamental importance of understanding ocean physics, chemistry, and circulation patterns in acidification research (Révelard et al., 2022; Robinson et al., 1999). This discipline provides essential knowledge regarding carbonate chemistry dynamics, pH variability, and the spatial-temporal patterns of acidification across different ocean basins (Osborne et al., 2022; Wu et al., 2018; Zeebe, 2012). The substantial presence of “Marine & Freshwater Biology” indicates the field’s strong biological foundation, encompassing research on organism-level responses, community dynamics, and ecosystem functioning (Flynn et al., 2012; Johnson et al., 2019). The connection between marine and freshwater biology suggests that acidification research extends beyond oceanic systems to include coastal and estuarine environments where freshwater-marine interaction create complex pH gradients (Kerrison et al., 2011; Lowe et al., 2019).

The network structure reveals sophisticated patterns of disciplinary integration. “geosciences”, “multidisciplinary” serves as a critical bridging discipline, connecting earth system science approaches with biological and chemical perspectives (Fennel et al., 2022). This positioning reflects the recognition that OA represents a global geochemical perturbation requiring earth system-level understanding (Steffen et al., 2016). “Ecology” node occupies a central position within the biological sciences cluster, indicating its role in organizing knowledge about community-level and ecosystem-level responses to acidification. The strong connections between ecology and other biological disciplines suggest that ecological theory provides

important conceptual frameworks for understanding how acidification cascades through marine food webs and alters ecosystem functioning(Jin et al., 2020; Zunino et al., 2021). The presence of “MICROBIOLOGY” as a distinct discipline highlights the specialized knowledge domain focused specifically on microbial responses to environmental change. Zhou et al. conducted manipulative experiments to assess the physiological and metabolic responses of nitrifying bacteria to OA under perturbations from anthropogenic nitrogen inputs. The results indicate that increased ammonium loading in estuarine and coastal waters mitigated the inhibitory effect of acidification on nitrification rates; however, oxidation of hydroxylamine by nitrifying bacteria enhanced the production of nitrous oxide, thereby potentially accelerating global climate change(Zhou et al., 2024). Its positioning within the broader biological sciences network indicates successful integration of microbiological research with community ecology and ecosystem science approaches(Liu et al., 2010).

The network structure demonstrates successful interdisciplinary synthesis, with multiple pathways connecting different knowledge domains. This pattern reflects the field’s evolution from chemistry-driven predictions toward organism-level validation, but the weakening peripheral nodes (Limnology, Atmospheric Sciences) indicate critical blind spots in understanding. The limited integration between limnology and core marine science disciplines suggests insufficient attention to freshwater-marine continuum effects in estuarine systems despite their role as microbial biodiversity hotspots(Crump & Bowen, 2024). Similarly, atmospheric CO₂ variability, where limited integration with climate modeling communities may lead to oversimplified experimental scenarios that fail to capture realistic environmental forcing(Kolby Smith et al., 2016).

3.3 Journal Co-citation Network Analysis

The journal co-citation network analysis reveals the intellectual structure and disciplinary foundations of OA and microbial ecology research(Fig.3). The network visualization demonstrates distinct clustering patterns that reflect the interdisciplinary nature of this research field, with clear connections between marine biology, oceanography, climate science, and biogeochemistry journals. The central position of core journals such as *NATURE*, *SCIENCE*, and *BIOGEOSCIENCES* indicates their foundational role in establishing the theoretical framework for OA research. These high-impact journals serve as primary knowledge sources, with *NATURE* and *SCIENCE* providing seminal papers that define the field’s conceptual boundaries, while *BIOGEOSCIENCES* contributes specialized research on biogeochemical processes(Ferderer et al., 2024). The prominence of *LIMNOL OCEANOGR* (Limnology and Oceanography) reflects the field’s strong oceanographic foundation, emphasizing the importance of physical and chemical oceanography in understanding acidification processes(Velthuis et al., 2022).

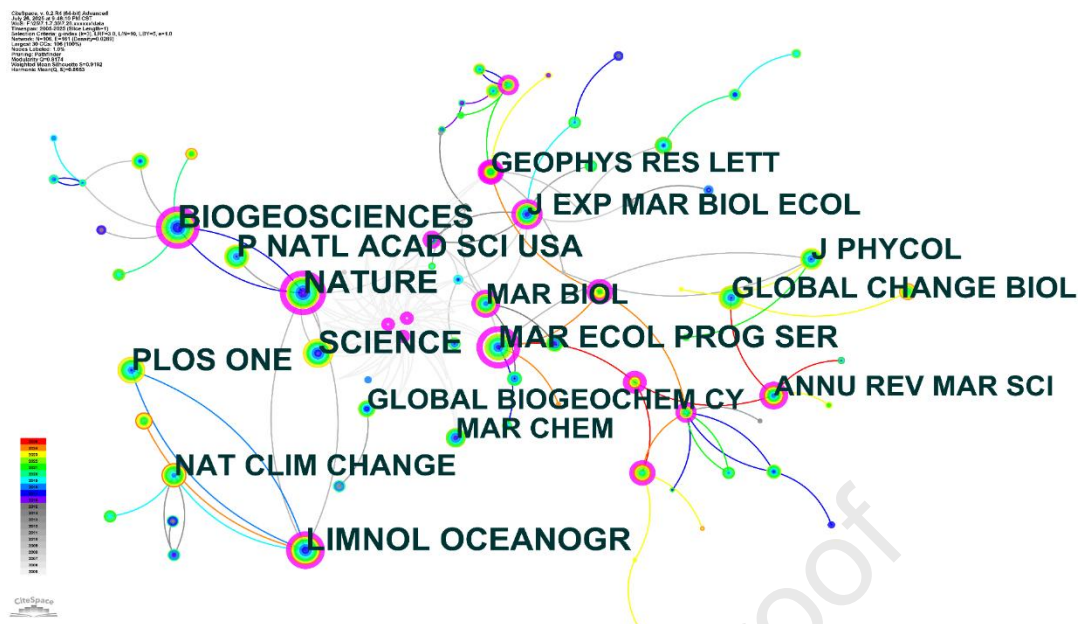


Figure 3. Journal co-citation network of acidification-microbial interaction research. Node ring colours represent the temporal distribution of publications or citations: cooler colours (blue, green) correspond to earlier years, while warmer colours (yellow, orange, red) indicate more recent activity. Purple rings signify high betweenness centrality.

The network structure reveals several distinct research clusters. The marine ecology cluster, anchored by journals such as *MAR BIOL* (Marine Biology), *MAR ECOL PROG SER* (Marine Ecology Progress Series), and *EXP MAR BIOL ECOL* (Experimental Marine Biology and Ecology), represents the biological response component of acidification research. This cluster focuses on organism-level and community-level effects, encompassing experimental studies on microbial physiology, community dynamics, and ecosystem responses. The strong connections between these journals indicate substantial knowledge exchange regarding biological impacts and experimental methodologies. The oceanographic and chemical cluster, including *GEOPHYS RES LETT* (Geophysical Research Letters) and *MAR CHEM* (Marine Chemistry), emphasizes the fundamental role of ocean chemistry and physics in acidification research (Uchikawa & Zeebe, 2008). These journals contribute essential knowledge regarding carbonate chemistry, pH dynamics, and the physical processes that govern acidification patterns across different ocean basins and depth layers (Monteiro et al., 2025; Qiu et al., 2023).

The climate and global change cluster, represented by journals including *GLOBAL CHANGE BIOL* (Global Change Biology), *NAT CLIM CHANGE* (Nature Climate Change), and *GLOBAL BIOGEOCHEM CY* (Global Biogeochemical Cycles), demonstrates the field's integration with broader climate science research. This clustering pattern reflects the recognition that OA represents a critical component of global environmental change, requiring interdisciplinary approaches that combine climate modeling, biogeochemical cycling, and ecological research (Olson et al., 2025; Tagliabue et al., 2011; Vodopivec, 2024).

The temporal dynamics reflected in the network's color coding indicate the evolution of research focus over time. Earlier research, represented by more traditional oceanographic and marine biology journals, established the foundational understanding of ocean chemistry and biological systems. More recent publications, indicated by warmer colors in the network, show increased representation in climate science and global change journals, reflecting the field's evolution toward broader environmental and policy relevance.

The presence of specialized journals such as J “*PHYCOL*” (Journal of Phycology) and “*ANNU REV MAR SCP*” (Annual Review of Marine Science) indicates the field's development of specialized knowledge domains while maintaining connections to broader scientific literature. The positioning of *PLOS ONE* within the network reflects the emergence of open-access journals as major nodes in the network represents an important shift in knowledge dissemination patterns, potentially increasing accessibility of research findings to broader audiences.

3.4 Literature Co-citation Network Analysis

The literature co-citation network (Fig.4) reveals three primary knowledge clusters that represent the foundational intellectual pillars of OA research. To ensure that the narrative cluster descriptions faithfully represent the co-citation structure visible in Figure 4, we cross-referenced the three clusters with the top-cited references in the co-citation network. The first cluster, anchored by Kroeker et al. (2013; 37 co-citations)(Kroeker et al., 2013b), Riebesell et al. (2013; 28 co-citations)(Aberle et al., 2013), Riebesell et al. (2007; 26 co-citations)(Riebesell et al., 2007), and Orr et al. (2005; 24 co-citations, centrality = 0.69)(Orr et al., 2005), was designated “Meta-analytic synthesis and foundational OA frameworks,” as it centers on large-scale reviews and landmark papers that established the theoretical and empirical baselines for OA biological impacts. The second cluster, anchored by Iglesias-Rodriguez et al. (2008; 34 co-citations) (Iglesias-Rodriguez et al., 2008) and Beaufort et al. (2011; 21 co-citations)(Beaufort et al., 2011), was designated “Coccolithophore calcification and experimental carbonate chemistry,” reflecting its concentration on calcifying phytoplankton responses and CO₂ manipulation methodologies. The third cluster, anchored by Schulz et al. (2013; 32 co-citations) (Aberle et al., 2013) and Brussaard et al. (2013; 28 co-citations)(Brussaard et al., 2013), was designated “Mesocosm experimental design and plankton community responses,” as it comprises papers employing large-scale CO₂ enrichment approaches to examine whole-community effects. These validated thematic labels are used throughout the subsequent discussion.

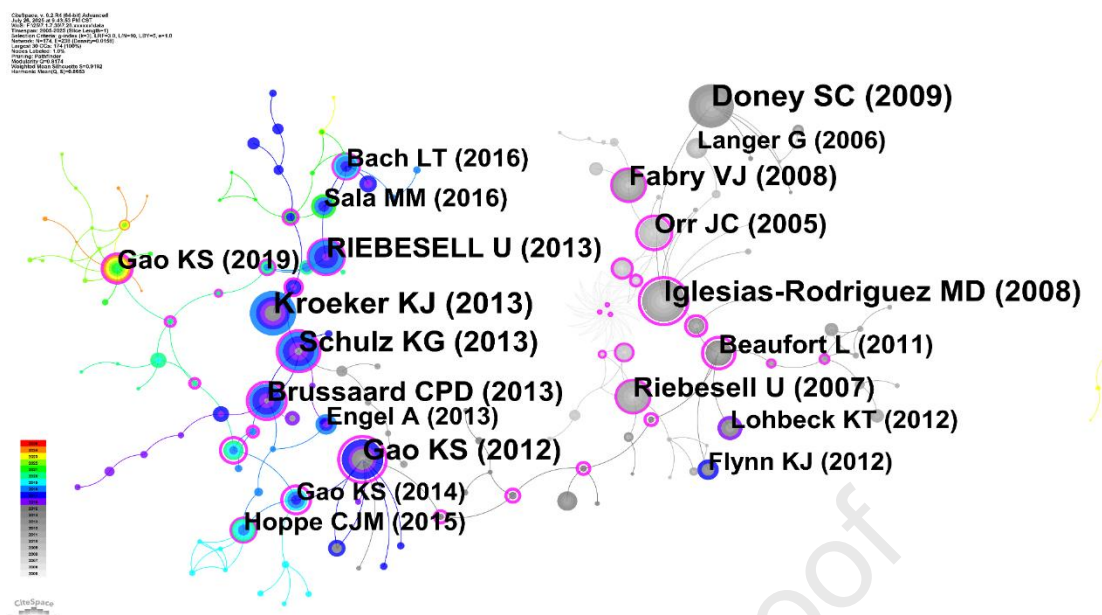


Figure 4. Literature co-citation network for acidification-microbial interaction. Node ring colours represent the temporal distribution of publications or citations: cooler colours (blue, green) correspond to earlier years, while warmer colours (yellow, orange, red) indicate more recent activity. Purple rings signify high betweenness centrality.

The temporal evolution visible in the network structure illuminates how research focus has shifted from early observational studies to sophisticated experimental approaches and ecosystem-level investigations. Early influential papers from 2005-2008, including Doney et al. (2009) (Doney, Fabry, et al., 2009b) and Fabry et al. (2008) (Fabry et al., 2008), established the basic chemical and biological principles of OA, forming dense co-citation relationships that persist throughout the network. The period from 2011-2013 represents a consolidation phase, with major synthesis works like Gao et al. (2012, 2014) (Gao & Campbell, 2014; Kunshan Gao et al., 2012) and multiple Riebesell publications creating a highly interconnected central hub that subsequent research consistently references. More recent studies from 2015-2019, including Bach et al. (2016, 2017) (Anil et al., 2016; Dam et al., 2017) and various specialized investigations, show increasingly diverse citation patterns, suggesting field maturation with specialized sub-disciplines emerging around specific organisms, experimental techniques, and ecosystem components. The network topology indicates that while foundational papers maintain their centrality, newer research is creating more specialized citation clusters, reflecting the field's evolution from broad theoretical understanding to detailed mechanistic investigations of specific microbial communities and biogeochemical processes under acidification scenarios (Currie et al., 2017; Nawaz et al., 2022).

The structural characteristics of the co-citation network further reveal the interdisciplinary nature of OA research and its integration with broader marine science disciplines. The positioning of key methodological papers, such as those by Schulz et al. (2013) (Schulz et al.,

2013) and Brussaard et al. (2013) (Brussaard et al., 2013) with 32 and 28 citations respectively, indicates strong methodological standardization within the field, with these works serving as essential references for experimental design and data interpretation. Both studies implemented a CO₂ enrichment protocol that increased dissolved inorganic carbon while keeping total alkalinity constant, thereby reproducing the chemical essence of OA. Depth-distributed injection using a dispersal device ensured uniform perturbations across the water column and avoided the TA artifacts often introduced by bubbling methods. In the same experimental arena, the authors established a continuous pCO₂ gradient spanning 185-1420 µatm, applied stepwise over several days. This enabled dose-response inference within a single, coeval environmental context and favored regression-based over groupwise statistics, thereby improving power and robustness.

The network's hub-and-spoke configuration around central nodes suggests that while the field has diversified into specialized research areas, it maintains strong theoretical coherence through consistent reference to foundational works. This pattern is particularly evident in the clustering of papers addressing specific taxonomic groups or experimental approaches, which nonetheless maintain strong co-citation relationships with the core theoretical literature, ensuring conceptual continuity across diverse research applications and maintaining the field's intellectual unity despite increasing specialization.

3.5 The Global Architecture of Scientific Production and Collaboration

The country collaboration network (Fig.5) consists of 53 countries presenting a globally distributed yet hierarchically structured research landscape. The network is dominated by China, USA, and Germany, represented by the largest nodes indicating highest publication output. The USA and Germany exhibit high betweenness centrality (indicated by purple rings), suggesting they play pivotal roles as bridges in international information flow. Following closely are the UK, France, Australia, and Norway, forming a mature research core.

The multi-colored citation rings on the "PEOPLES R CHINA" node, with strong emphasis on recent years (yellow and orange outer rings), point to rapid emergence as a major research power. China shows growing connections with countries including Canada and Argentina, indicating expanding strategic engagement. However, compared to the USA and Germany, China's network position suggests it operates primarily through bilateral partnerships rather than as a fully independent network hub.



Figure 5. Country co-occurrence map for acidification-microbial interaction research.

While the network spans 53 countries, substantial regional gaps exist. The bibliometric data reveals limited representation from Africa, South America (excluding major research institutions in Brazil), Southeast Asian developing nations, and Small Island Developing States. These regions face significant OA threats but remain underrepresented in the global research network. To quantify this imbalance, we examined the 53-country co-authorship network. African nations contributed only 6 papers across three countries: Egypt (3), South Africa (2), and Morocco (1). Southeast Asian institutions were similarly limited, with Malaysia (4), Singapore (2), and the Philippines (1) as the sole participating nations. No Small Island Developing States appear in the network at all. This pattern likely reflects a combination of factors: limited local research infrastructure and funding, but also systemic biases inherent in our analytical approach. The exclusive reliance on Web of Science, which favors English-language journals indexed by major publishers, may undercount research published in regional journals, non-English languages, or grey literature from developing nations (Mongeon & Paul-Hus, 2015). For Small Island Developing States in particular, underrepresentation is not merely a capacity issue but reflects broader patterns of knowledge inequity in global environmental science, where nations most vulnerable to climate-driven ocean change often have the least voice in shaping the research agenda (Karlsson et al., 2007; Pasgaard et al., 2015). Addressing this imbalance will require not only capacity-building investments but also deliberate inclusion of Global South researchers in international collaborative networks and co-design of research priorities.

The institutional co-citation network analysis (Fig.6) reveals the global research infrastructure and collaborative patterns underlying OA and microbial ecology research. The network visualization demonstrates a well-established international research community characterized

by strong European leadership, emerging Asia-Pacific contributions, and strategic institutional partnerships that span multiple continents.

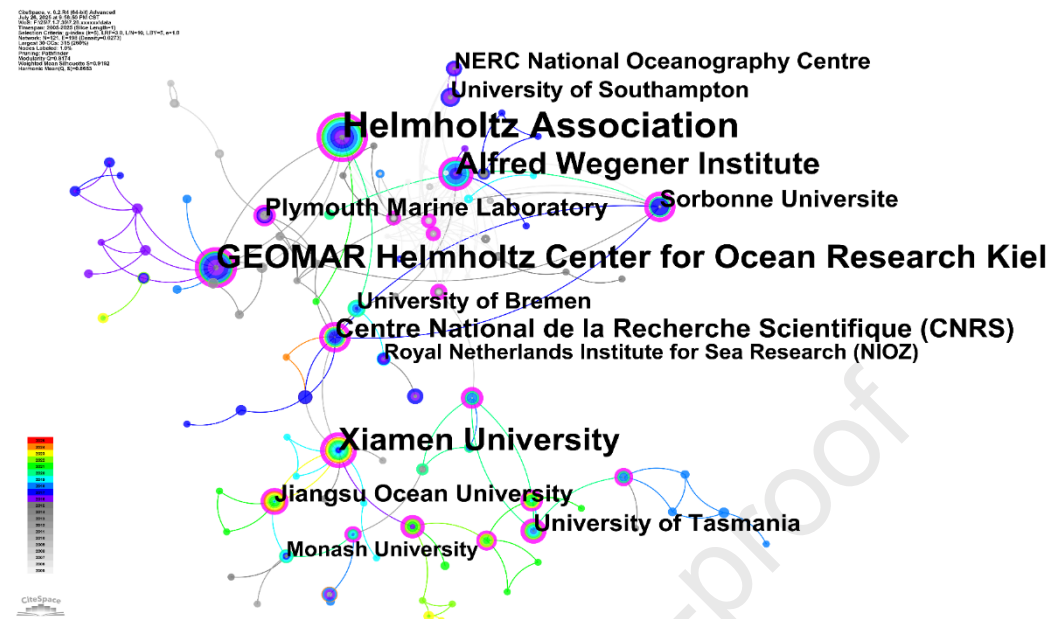


Figure 6. Institution co-occurrence map for acidification-microbial interaction research.

The network structure clearly demonstrates European dominance in OA and microbial ecology research. GEOMAR Helmholtz Center for Ocean Research Kiel (66 papers) occupies the most central position in the institutional network. This centrality is consistent with GEOMAR's operational leadership of the KOSMOS mesocosm platform, which has hosted many of the large-scale CO₂ perturbation experiments that generated highly cited publications in this dataset (Anil et al., 2016; Riebesell et al., 2013; Schulz et al., 2013). The institution's role as coordinator of multiple EU-funded projects (e.g., BIOACID, EPOCA) further explains its bridging function in the collaboration network, as these programs required multi-national partnerships that are directly reflected in the co-authorship links visible in Figure 6. However, that network centrality captures collaborative connectivity rather than research quality per se.

The institution's strong position suggests it serves as a critical knowledge hub, facilitating collaboration and knowledge exchange across the international research community. The Helmholtz Association (106 papers) appears as a distinct network node, highlighting the important role of this German research organization in coordinating large-scale, interdisciplinary research efforts. The Helmholtz Association's involvement indicates that OA research benefits from substantial institutional support for long-term, coordinated research programs that transcend individual laboratory capabilities. The Alfred Wegener Institute (52 papers) represents another major German contribution to the field, particularly renowned for polar and marine research (Ahme et al., 2023; Jenrich et al., 2024). Its position in the network reflects the critical importance of polar regions in global acidification research, as these environments experience particularly rapid acidification and serve as sentinel systems for

understanding global change impacts on marine ecosystems (Campen et al., 2021; Rees et al., 2021; Wang et al., 2023).

In addition, Plymouth Marine Laboratory (21 papers) in the United Kingdom occupies a strategic position, bridging European continental research with UK-based institutions. The Centre National de la Recherche Scientifique (CNRS, 27 papers) represents France's substantial contribution to the field, indicating the importance of the French national research system in advancing OA science. The node of CNRS suggests strong integration between French research institutions and the broader European research network, facilitating knowledge transfer and collaborative research initiatives.

Meanwhile, the network reveals significant Asia-Pacific research contributions, indicating the global expansion of acidification research capacity. Xiamen University (57 papers) occupies a prominent position, representing China's growing leadership in marine science research. The university's centrality suggests it serves as a major hub for acidification research in the Asia-Pacific region, facilitating regional collaboration and contributing unique perspectives on acidification impacts in tropical and subtropical systems. The positioning of Chinese institutions suggests successful integration with international research networks while developing independent research capabilities. Monash University (13 papers) and the University of Tasmania (18 papers) represent Australia's significant contributions to acidification research. Tasmania's prominence is particularly notable given its strategic location in the Southern Ocean, where rapid acidification intersects with unique marine ecosystems (Trull et al., 2018). These Australian institutions contribute essential knowledge regarding acidification impacts in the Southern Hemisphere and on cold-water marine communities. The color rings of nodes suggest evolution from European-dominated research toward more globally distributed capacity, with earlier research phases showing strong European foundations while more recent developments indicate expanded Asia-Pacific participation and emerging leadership.

3.6 Influential Authors Network Analysis

The comprehensive analysis of author networks in OA impacts research on microbial ecology reveals both the collaborative structure of the research community and the intellectual foundations that have shaped the field's development. The author collaboration network (Fig.7) consisting of 493 authors reveals well-established research community with several prominent clusters. The high modularity ($Q = 0.8174$) and mean silhouette score ($S = 0.9192$) indicate the network is clearly divided into distinct and highly coherent research groups. The most prolific author is Ulf Riebesell, followed by Kunshan Gao, Lennart T. Bach, and John Beardall, who also serve as important network bridges.

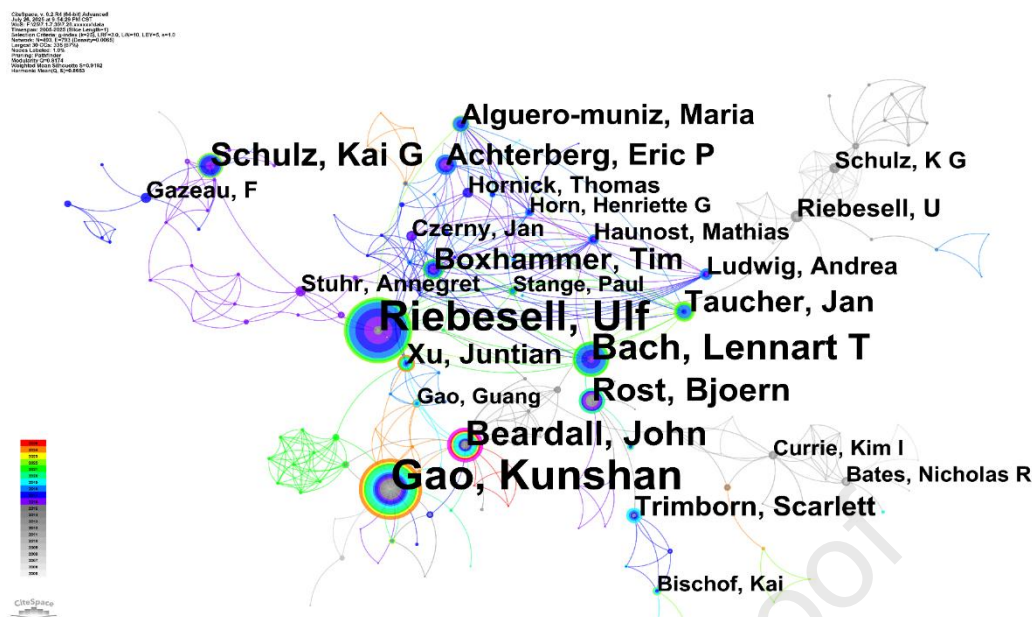
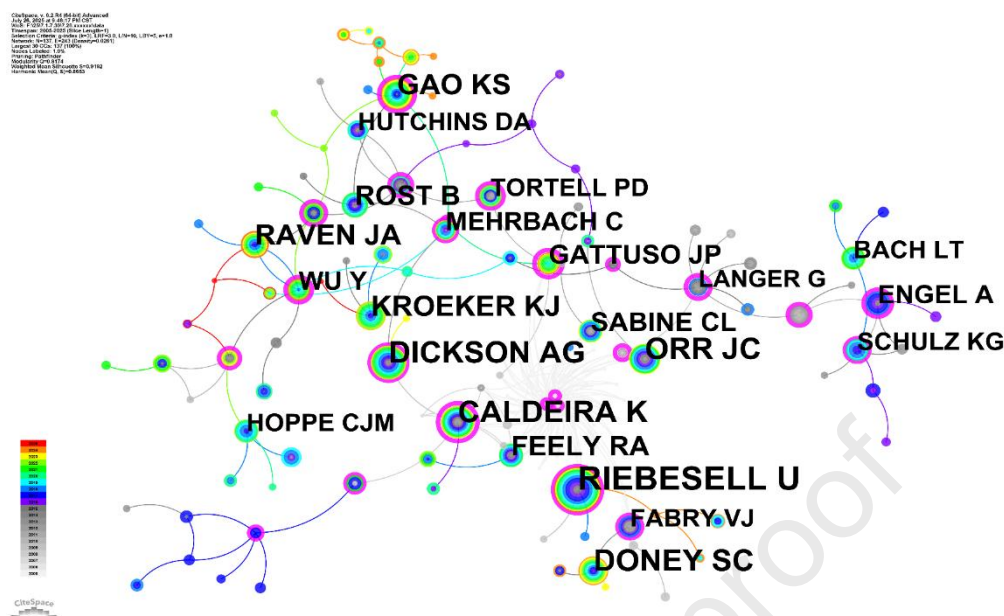


Figure 7. Author co-occurrence map for acidification-microbial interaction research. Node ring colours represent the temporal distribution of publications or citations: cooler colours (blue, green) correspond to earlier years, while warmer colours (yellow, orange, red) indicate more recent activity. Purple rings signify high betweenness centrality.

Meanwhile, Ulf Riebesell, Kunshan Gao and John Beardall et al. also serve as important bridges within the network. Analysis of the primary clusters reveals the key collaborative powerhouses in this research area. The largest and most central cluster is a tightly-knit group led by Ulf Riebesell, with core members including Lennart T. Bach, Tim Boxhammer, Kai G. Schulz, and Eric P. Achterberg. The color rings of these nodes, spanning from blue to vibrant green and yellow, demonstrate this group's long-standing and sustained contributions to the field. A second major cluster is centered around Kunshan Gao, who frequently collaborates with John Beardall and Juntian Xu. The bright, warm-colored rings of Gao's node signify a significant volume of high-impact research in more recent years. Other notable, albeit smaller, clusters include a group associated with Fabrice Gazeau and another involving Nicholas R. Bates and Kim I. Currie. The color links indicate that while foundational collaborations were established by these core groups, the field has seen dynamic expansion of its collaborative web, integrating new researchers and fostering inter-cluster connections.

The co-citation network of authors delineates the intellectual foundation and the primary research fronts of the field (Fig.8). In contrast, the author co-citation cluster map emphasizes scholarly impact rather than collaboration (X. Zhang et al., 2025). The network comprising 136 nodes with $Q = 0.8174$ and $S = 0.9192$ delineates the intellectual foundation of the field. The most prominent nodes belong to U. Riebesell, K. Caldeira, Gao KS, J.C. Orr, J.-P. Gattuso, and S.C. Doney, indicating their status as the most highly cited and foundational authors. Several key authors (K. Caldeira, Gao KS, J.-P. Gattuso) exhibit highest betweenness centrality,

586 signifying their work serves pivotal bridging functions connecting different sub-fields.



587
588 **Figure 8. Author co-citation map for acidification-microbial interaction research. Node ring colours**
589 **represent the temporal distribution of publications or citations: cooler colours (blue, green) correspond to**
590 **earlier years, while warmer colours (yellow, orange, red) indicate more recent activity. Purple rings signify**
591 **high betweenness centrality.**

592 A large central cluster comprising foundational authors (Caldeira, Feely, Orr, Gattuso, Riebesell,
593 Doney) represents the core intellectual domain focused on fundamental biogeochemistry and
594 global-scale impacts. A second significant cluster includes K. S. Gao, D. A. Hutchins, B. Rost,
595 and J. A. Raven, suggesting a research front concentrated on the physiological and ecological
596 responses of phytoplankton and primary producers to changing ocean chemistry(K. Gao et al.,
597 2012; Hutchins et al., 2023; Raven & Crawford, 2012; Rost et al., 2008). Another distinct group,
598 featuring L. T. Bach, A. Engel, and K. G. Schulz, points to a specialized focus, possibly on the
599 role of specific plankton groups like coccolithophores and their influence on biogeochemical
600 cycles(Aberle et al., 2013; Bach et al., 2011; Dam et al., 2017). Notably, the small but central
601 cluster of A. G. Dickson and C. Mehrbach highlights the critical importance of their
602 foundational work on seawater carbonate system measurements, which underpins the
603 methodological consistency of the entire field(Carter et al., 2023). The temporal aspect,
604 illustrated by the multi-colored rings on nodes like Riebesell and Caldeira, shows their
605 sustained citation impact over the entire study period, confirming the long-term relevance of
606 their foundational contributions.

607 3.7 Keyword Co-occurrence: Conceptual Architecture

608 The keyword co-occurrence network (Figure 9) reveals sophisticated conceptual architecture
609 with “ocean acidification” serving as the dominant central hub exhibiting largest node size and
610 highest connectivity. The positioning of core concepts such as “climate change,” “CO₂,”

experiments show that elevated $p\text{CO}_2$ can down-regulate CCM investment in some taxa/communities (Brandenburg et al., 2019; Young et al., 2015), yet the realized growth benefits are often modest because CCM savings can be counterbalanced by other costs and constraints; multi-stressor contexts (warming, light, nutrients) frequently dominate net outcomes (Gao et al., 2019a). Moreover, recent work demonstrates temperature sensitivity of CCMs in diatoms, implying interactive controls that help explain species- and season-specific responses (Li & Young, 2023). Collectively, these results argue that CCM plasticity is a necessary but insufficient predictor of OA responses unless embedded in the full resource and thermal context.

The temporal color coding within these clusters suggests that while early research focused on broad organismal responses, more recent investigations have delved into specific cellular and molecular mechanisms, reflecting the field's progression toward mechanistic understanding of acidification effects on microbial communities and their biogeochemical functions (Liang et al., 2023; Simonetti et al., 2022). An integrative analysis combining metabolomics and 16S rRNA amplicon sequencing elucidated the underlying mechanisms by which *Macrocystis pyrifera* and its symbiotic bacteria respond to OA. OA altered the structure of the *M. pyrifera*-associated microbiome, shifted its metabolic profile, and reshaped microbe-metabolite interactions. Stress-responsive molecules—including fatty-acid metabolites, oxylipins, and hormone-like compounds—emerged as key metabolomic indicators (Zhang et al., 2024).

The positioning of terms such as “temperature,” “impacts,” “responses,” and “dynamics” in intermediate network positions suggests their role as bridging concepts connecting fundamental acidification processes with broader ecological contexts. The integration of “diversity” and “inorganic carbon” indicates research has expanded beyond single-species studies to encompass community-level responses and biogeochemical cycling implications (Currie et al., 2017; Hassenrück et al., 2016; Iida et al., 2020).

However, a notable absence in the keyword co-occurrence network deserves attention. To validate this observation beyond keyword frequency, we conducted a supplementary search within the same 495-paper dataset. Searching for “archaea,” “archaeal,” or “thaumarchae” in titles, abstracts, and keywords returned only 9 papers (1.8% of the corpus). A parallel search for “virus,” “viral,” “phage,” or “virome” returned 10 papers (2.0%). Furthermore, neither “archaea” nor any virus-related term appears among the 217 nodes in the keyword co-occurrence network, indicating that these topics have not reached sufficient publication volume to form identifiable conceptual nodes in the field's knowledge structure. These proportions are strikingly low given the ecological importance of these groups: archaea dominate ammonia oxidation in many marine environments and mediate key steps in nitrogen and carbon cycling under low-pH conditions (Offre et al., 2013), while viruses regulate microbial population dynamics and organic matter turnover through lysis and lysogeny, processes demonstrably

altered by acidification (Tangherlini et al., 2021; Malits et al., 2021). The combined 19 papers represent less than 4% of the corpus, confirming that the absence of these taxa from high-frequency keyword clusters reflects a genuine thematic gap.

3.8 Keyword Clustering: Thematic Domains

The keyword clustering analysis (Fig.10) generated ten major thematic clusters with high modularity ($Q = 0.8174$) and mean silhouette value ($S = 0.9192$), indicating well-defined research areas. The largest cluster (#0) labeled “growth” encompasses studies investigating microbial growth responses to lowered pH and elevated pCO_2 (Deppeler et al., 2018; James et al., 2019; Newbold et al., 2012), often employing microcosms or mesocosms to assess physiological plasticity and shifts in productivity across diverse microbial taxa (Spilling et al., 2016; Spisla et al., 2021). For example, on-board pCO_2 perturbation experiments in the Eastern Tropical South Pacific demonstrated that even short-term exposure to elevated pCO_2 can significantly alter bacterial community composition, reducing SAR11 abundance and restructuring nitrogen-cycling taxa (P. Aguayo et al., 2020). Cluster #1 “elemental ratios” focuses on changes in carbon, nitrogen, and phosphorus stoichiometry, providing mechanistic understanding of altered biogeochemical cycling (Taucher et al., 2020).

Cluster #2 “ocean acidification” serves as a central thematic bridge linked to nearly all other clusters, reflecting its foundational role. Its proximity to clusters such as #3 (“climate change”) and #7 (“ CO_2 levels”) further highlights the convergence of acidification research with broader climate-related stressors. Cluster #4, “reduced calcification,” and cluster #5, “ N_2 fixation,” represent specialized but increasingly interconnected areas, as studies reveal how changes in carbonate chemistry and trace metal bioavailability under acidification scenarios may influence both calcifying organisms and diazotrophic microbial communities (Cohen et al., 2009; Stockdale et al., 2016). Cluster #6 “carbon” suggests cross-cutting interest in carbon cycling, encompassing carbon fixation, respiration, and dissolved organic carbon dynamics (Simone et al., 2021). Meanwhile, clusters #8 (“dissociation”) and #9 (“anthropogenic carbon”) represent more chemically or modeling-oriented subfields, emphasizing carbonate system equilibrium and anthropogenic CO_2 attribution, respectively (Deppeler et al., 2018; Heinze et al., 2015).

The clustering structure demonstrates a field that is both thematically diverse and intellectually integrated. The adjacency and interconnectivity of clusters reflect a growing trend toward interdisciplinary inquiry—linking microbial physiology with global change biology and Earth system processes. The clear boundaries between clusters such as “elemental ratios,” “ N_2 fixation,” and “reduced calcification” also suggest that while conceptual bridges are emerging, further interdisciplinary collaboration across subfields remains a promising direction for future research.

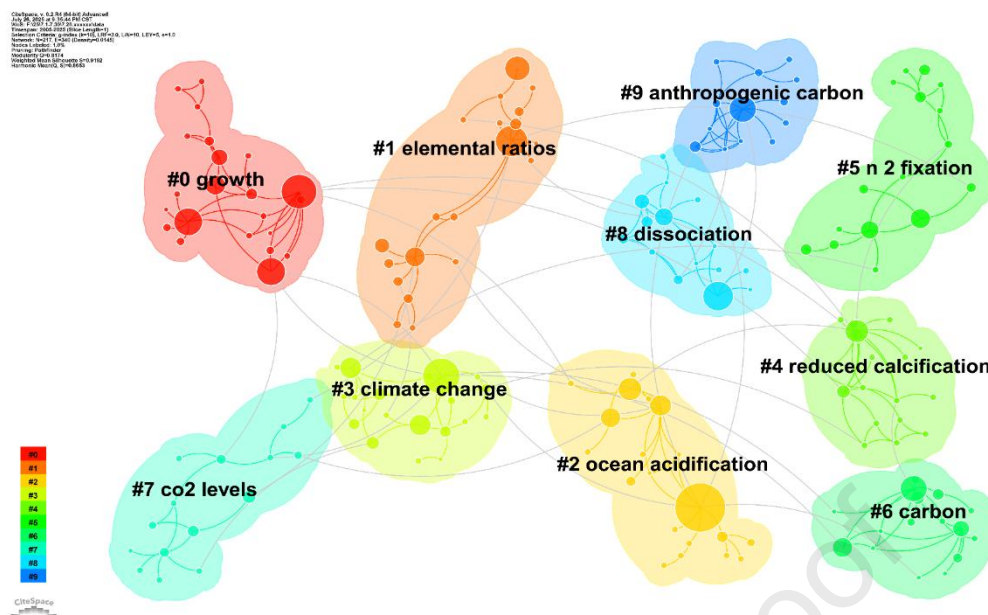


Figure 10. Co-occurring keyword cluster network.

3.9 Temporal Keyword Evolution: Thematic Transitions

The timeline visualization of keyword emergence reveals distinct phases of conceptual evolution in OA research, demonstrating a clear progression from foundational chemical concepts to sophisticated biological and ecological investigations spanning two decades (Fig.11). The early foundational period (2005-2010) is characterized by the establishment of core chemical and physical parameters, with fundamental terms such as “CO₂,” “carbon dioxide,” “ocean acidification,” and “southern ocean” appearing as the initial conceptual framework. This period represents the essential groundwork where researchers established the basic chemistry of OA, with “calcium carbonate” and “calcification” emerging as critical early concepts that linked atmospheric CO₂ increases to direct impacts on marine calcifying organisms (Hofmann et al., 2010). The positioning of “seawater” and “climate change” during this phase indicates that early research focused on understanding the basic mechanistic relationships between global carbon cycling and ocean chemistry changes (Doney, Balch, et al., 2009; Dore et al., 2009), establishing the environmental context that would drive all subsequent microbiology investigations.

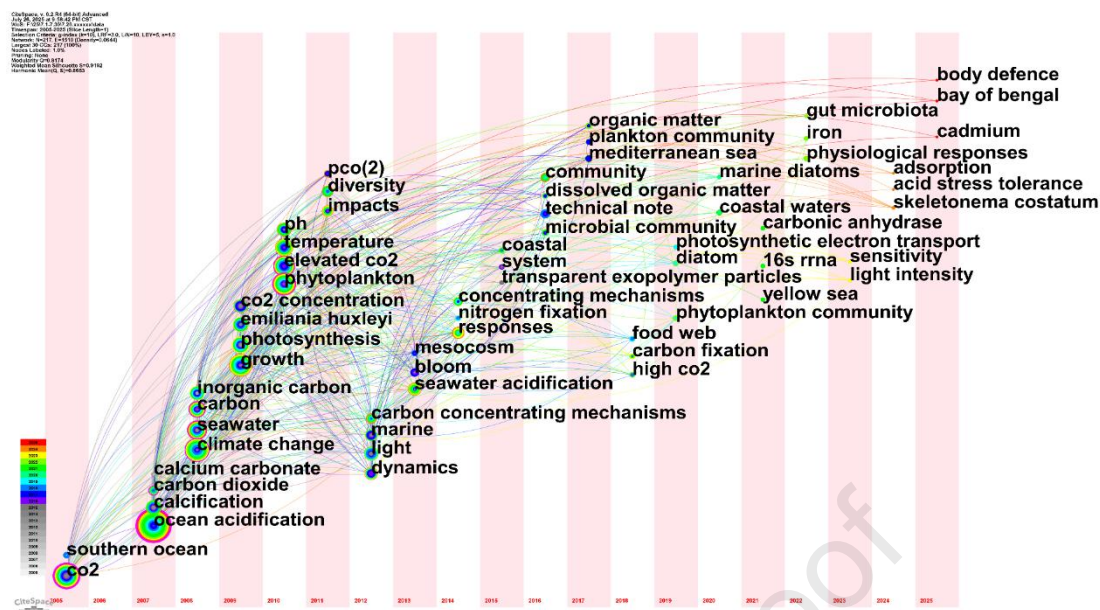


Figure 11. Time zone view of keywords of acidification-microbial interaction.

The intermediate expansion phase (2011-2016) demonstrates substantial thematic diversification as research transitioned from basic chemistry to biological responses and ecological processes. The emergence of organism-specific terms such as “*Emiliana huxleyi*” and “phytoplankton” alongside process-oriented concepts like “photosynthesis,” “growth,” and “carbon concentrating mechanisms” reveals how the field evolved to examine specific biological responses to acidification stress (Bach et al., 2011; Lohbeck et al., 2012). The appearance of methodological terms including “elevated CO₂,” “CO₂ concentration,” and “inorganic carbon” during this period indicates increasing experimental sophistication, with researchers developing standardized approaches to manipulate and measure acidification effects (Cornwall & Hurd, 2016; Kroeker et al., 2013b). The temporal clustering of “mesocosm,” “bloom,” and “responses” suggests that this phase was characterized by scaled experimental approaches that moved beyond simple laboratory studies to more realistic ecological simulations, representing a critical methodological advancement that enabled more comprehensive understanding of acidification impacts on marine ecosystems (Anil et al., 2016; Zark et al., 2015a).

The recent specialization phase (2017-2025) reveals remarkable thematic diversification and increasing integration with broader marine ecology and environmental science disciplines. The emergence of highly specialized terms such as “16s rna,” “carbonic anhydrase,” and “photosynthetic electron transport” indicates that research has progressed to molecular and cellular levels, examining specific physiological and biochemical mechanisms underlying acidification responses (Ianora et al., 2017; McNicholl et al., 2019). A recent study reported that the marine bacterium *Pseudomonas sihuiensis* -BFB-6S adapts to the adverse effects of OA by modulating extracellular polymeric substances (EPS). Variations in OA levels modify the EPS

matrix, altering the total contents of proteins, carbohydrates, and lipids, as well as fluorophore composition and the relative abundance of aromatic amino acids such as tryptophan and tyrosine(Chakraborty et al., 2024b). The appearance of ecosystem-level concepts including “food web,” “community,” “microbial community,” and “plankton community” demonstrates concurrent scaling up to understand broader ecological implications beyond individual organism responses(Barreto et al., 2021). The integration of environmental context terms such as “mediterranean sea,” “bay of bengal,” “coastal waters,” and “yellow sea” reflects increasing recognition of regional variability and ecosystem-specific responses to acidification. The recent emergence of interdisciplinary concepts like “acid stress tolerance,” “physiological responses,” and “adsorption” suggests that the field is expanding to incorporate stress biology, environmental chemistry, and biogeochemical cycling perspectives, indicating maturation into a comprehensive interdisciplinary research domain that addresses acidification impacts across multiple biological scales and environmental contexts(Barrett et al., 2025; Kroeker et al., 2017).

3.10 Citation Bursts: Identifying Research Frontiers

The citation burst analysis reveals distinct phases of research intensity and paradigm shifts in OA studies, characterized by temporal patterns of keyword emergence that reflect the field’s evolution from fundamental chemical understanding to complex biological and ecological investigations (Fig.12). The early burst period demonstrates strong research focus on foundational biogeochemical processes, with “calcification” (strength 4.76, 2007-2012) and “calcium carbonate” (strength 4.45, 2007-2011) representing the initial core concerns about direct impacts of acidification on calcifying organisms. Mean calcification rates tend to decline under elevated pCO_2 , but the magnitude and even direction of response vary among clades and environments, reflecting differences in carbonate chemistry exposure, organismal control of the calcifying milieu, and microhabitat hydrodynamics (Harvey et al., 2013; Kroeker et al., 2013a; Kroeker et al., 2010). Mechanistically, scleractinian corals exemplify internal regulation of the calcifying fluid pH (pH_{cf}); spatial variability in pH_{cf} within colonies and species-level capacity to up-regulate pH_{cf} help explain heterogeneous calcification sensitivities to OA and its interaction with warming (Holcomb et al., 2014; McCulloch et al., 2017). At the organism-environment interface, diffusive boundary layers (DBLs) can buffer coralline algae from external pH excursions under low flow, underscoring that exposure at the scale actually experienced by the skeleton (not bulk seawater alone) is predictive of outcome(Cornwall et al., 2014). For coccolithophores, responses range from modest changes in calcite mass to malformed coccoliths, and are notably amplified by multi-stressor contexts, indicating that predictions for pelagic carbon export must incorporate OA - warming and OA - salinity interactions rather than OA alone(D’Amario et al., 2020; Xu et al., 2020).

The concurrent emergence of “inorganic carbon” as the strongest early burst (strength 7.05, 2008-2012) indicates intensive research efforts to understand carbon chemistry dynamics in

seawater, while “photosynthesis” (strength 4.84, 2009-2012) reflects early recognition that acidification affects not only calcification but also primary productivity. The appearance of “anthropogenic CO₂” (strength 3.15, 2008-2009) and “dissociation” (strength 6.28, 2010-2014) during this period establishes the human causation framework and chemical mechanistic understanding that became fundamental to the field, with “CO₂ concentrating mechanisms” (strength 3.22, 2010-2015) indicating early investigations into physiological adaptations of marine microorganisms to changing carbonate chemistry (Kroeker et al., 2017).

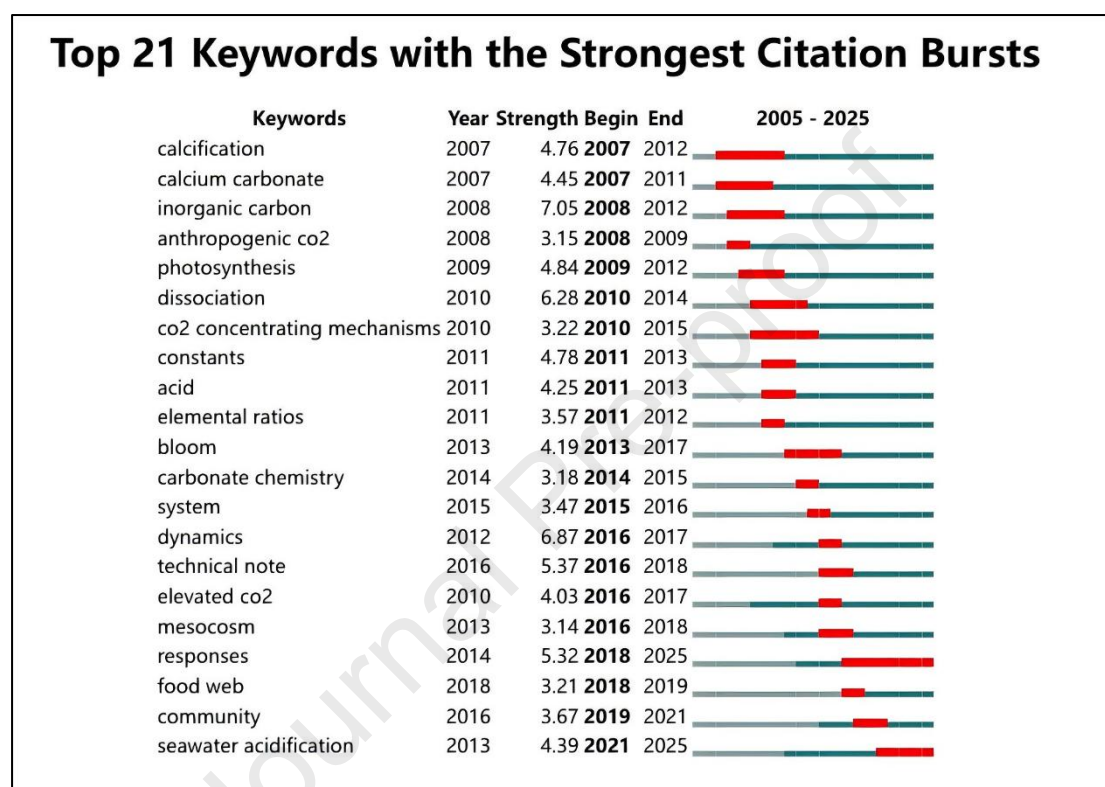


Figure 12. Top 21 keywords with the strongest citation bursts (2005-2025).

The intermediate methodological intensification phase reveals a shift toward experimental sophistication and ecological scaling, characterized by bursts in methodological and analytical approaches that enabled more comprehensive investigations of acidification impacts. The emergence of “constants” (strength 4.78, 2011-2013) and “acid” (strength 4.25, 2011-2013) reflects intensification of chemical measurement standardization, while “elemental ratios” (strength 3.57, 2011-2012) indicates increasing attention to biogeochemical stoichiometry under acidification stress. The delayed but strong burst of “bloom” (strength 4.19, 2013-2017) demonstrates growing interest in population-level responses and ecological phenomena, coinciding with “carbonate chemistry” (strength 3.18, 2014-2015) and “system” (strength 3.47, 2015-2016) which reflect more comprehensive ecosystem-level investigations. The notable temporal displacement of “dynamics” (strength 6.87, 2016-2017) and “elevated co₂” (strength 4.03, 2016-2017) from their initial appearance years suggests that these concepts gained renewed research intensity as experimental approaches matured, while “technical note”

(strength 5.37, 2016-2018) and “mesocosm” (strength 3.14, 2016-2018) indicate methodological advances in controlled experimental systems that enabled more realistic ecological simulations. Mesocosm studies report stimulated bacterial growth and altered enzyme activities under elevated $p\text{CO}_2$, even when bulk dissolved organic matter concentrations change little (Endres et al., 2014; Zark et al., 2015b). Molecular-level analyses further show that OA can modify biomolecule composition of planktonic organic matter (amino acids, carbohydrates), with consequences for particle aggregation, bacterial degradation, and the transfer of essential biomolecules to higher trophic levels (Bermúdez et al., 2016; Grosse et al., 2020). These results caution against using dissolved organic matter standing stocks as sole indicators and emphasize composition-sensitive metrics in OA experiments.

The contemporary research expansion phase demonstrates sustained research intensity in ecological and community-level investigations, with “responses” (strength 5.32, 2018-2025) showing the most prolonged and intense burst pattern, indicating that understanding biological and ecological responses remains the primary research focus. Because “responses” is an inherently broad term that could appear in virtually any environmental stress study, we conducted a manual content analysis of all 34 papers contributing to this burst during 2018–2025 to identify the specific research directions it encompasses. This analysis revealed three dominant sub-themes (papers may belong to multiple categories): (i) physiological and metabolic responses (23 papers, 68%), including altered photosynthetic performance, enzyme activities, extracellular polymeric substance production, and carbon/nitrogen fixation adjustments (Chakraborty et al., 2024a; James et al., 2019); (ii) community-structural responses (17 papers, 50%), including shifts in microbial diversity, species turnover, and functional group composition under elevated $p\text{CO}_2$ (Paulina Aguayo et al., 2020; Spisla et al., 2021); and (iii) multi-stressor interaction responses (16 papers, 47%), where OA effects are examined in combination with warming, UV radiation, hypoxia, or nutrient perturbation (Kaan Gundogdu et al., 2023; Hu et al., 2024). Notably, evolutionary and epigenetic responses (e.g., transgenerational plasticity, adaptive evolution) were represented by only 2 of the 34 papers (6%), suggesting that this sub-direction remains a largely unexplored frontier within the broader “responses” theme.

The sequential emergence of “food web” (strength 3.21, 2018-2019) and “community” (strength 3.67, 2019-2021) reflects the field’s evolution toward ecosystem-level understanding, examining how acidification cascades through marine ecological networks rather than focusing solely on individual organism responses. High biodiversity can buffer against the adverse effects of OA, primarily through increased food-resource availability and strengthened microbe-host symbioses (Rastelli et al., 2020). The late-period burst of “seawater acidification” (strength 4.39, 2021-2025) needs careful interpretation, since this term is broadly synonymous with “ocean acidification.” We examined whether this burst reflects a genuine research shift or

is an artifact of our search strategy, which included “seawater acidification” as a query term. First, CiteSpace’s burst detection operates on Keywords Plus and author keywords rather than title/abstract text, and only 8 papers in our dataset use “seawater acidification” as an author keyword, with 6 of 8 appearing after 2020. Second, the proportion of annual publications mentioning “seawater acidification” in any metadata field has increased from approximately 5-12% during 2016–2020 to 17-29% during 2021-2025, suggesting a real terminological shift rather than a simple indexing artifact. Third, among the 8 papers that explicitly chose “seawater acidification” as an author keyword, 5 (62%) address coastal, estuarine, or nearshore environments, compared with approximately 31% of the overall corpus. Taken together, these patterns suggest that the burst reflects an expanding research emphasis on coastal and estuarine acidification, where local processes such as riverine input, eutrophication-driven pH variability, and sediment–water interactions make “seawater” a more precise descriptor than “ocean.” Nevertheless, a partial indexing contribution cannot be fully excluded, and this interpretation should be treated as preliminary.

4. Prospects for OA and Microbial Ecology Research Field

The bibliometric patterns documented in Sections 3.1–3.10 converge on four structural features that point toward specific, evidence-linked research priorities. First, the keyword co-occurrence network (Section 3.7) reveals that molecular-level terms (“16S rRNA,” “carbonic anhydrase,” “metagenomics”) emerged only in the post-2017 phase and remain peripheral to the network core, indicating that advanced molecular technologies have not yet been systematically integrated into OA-microbe research, thus generating direction 1. Second, the geographic collaboration analysis (Section 3.5) documents severe spatial imbalances: 53 countries participate, but Africa, Southeast Asia, and Small Island Developing States remain under-represented despite facing acute OA threats, and even well-connected nations rely predominantly on short-term experimental designs, hence direction 2 on spatiotemporal expansion. Third, the bridging position of “temperature” and “dynamics” in the keyword network, combined with the late-emerging burst of “responses” (strength 5.32, 2018-2025), demonstrates that multi-stressor research is growing but still lags behind single-factor OA studies in citation volume, thus generating direction 3. Fourth, the disconnect between organism-level keyword clusters (#0 “growth,” #4 “reduced calcification”) and ecosystem-scale clusters (#3 “climate change,” #6 “carbon”) visible in the clustering map (Section 3.8) reveals a persistent scale-integration gap that demands simultaneous advances in experimental infrastructure, computational integration, and science-policy translation, thus generating direction 4. Below, we elaborate each direction with specific methodological recommendations grounded in these bibliometric findings.

4.1 Integration of Advanced Molecular Technologies and Systems Biology

The keyword co-occurrence network (Section 3.7) contains no molecular-technique terms such

as “metatranscriptomics,” “metagenomics,” or “single-cell genomics” among its 217 nodes, and the supplementary search revealed that archaea (9 papers) and viruses (10 papers) together account for less than 4% of the corpus (Section 3.7). This dual absence of both methodological keywords and taxonomic targets indicates that molecular-resolution investigation of OA–microbe interactions remains a pronounced gap in the current literature.

The temporal shift in keywords from carbonate chemistry to organismal and ecological terms foreshadows a methodological inflection point: integrated multi-omics at scale. The peripheral positioning of molecular technique keywords in the co-occurrence network (Section 3.7) suggests that, despite individual high-impact studies, the field has not yet routinely adopted multi-omics frameworks. Pairing metagenomes with metatranscriptomes, metaproteomes, and metabolomes will resolve regulatory architecture and metabolic plasticity that underpins microbial adaptation to declining pH(Wang et al., 2021). Critically, single-cell genomics now routinely recovers genomes from uncultured lineages and picoeukaryotes, exposing clade-specific acclimation strategies and within-population heterogeneity that aggregate statistics cannot capture. The taxonomic bias identified in our keyword analysis — where “*Emiliana huxleyi*” dominates organism-level nodes while archaea and viruses are absent — underscores the urgency of applying these technologies to under-studied lineages. This gap is particularly significant given emerging evidence that OA fundamentally alters virus–prokaryote dynamics. For instance, shifting viral strategies from lytic to chronic infection in benthic ecosystems(Tangherlini et al., 2021), yet such interactions remain invisible in the current bibliometric landscape. Incorporating these data streams into systems-level models will allow explicit tests of how cellular controls propagate to community- and ecosystem-level functions under acidification.

A second, rapidly maturing frontier is the fusion of AI/ML with multi-omics and imaging. Deep-learning pipelines trained on plankton image archives and molecular signatures already approach human-level classification in several contexts and can be adapted to detect OA-linked trait shifts, flag early-warning biomarkers, and forecast community restructuring under projected carbonate trajectories(Grosse et al., 2020). Purpose-built workflows and FAIR data practices will be essential to reproducibility and model transfer.

The taxonomic bias identified in the keyword network — where “*Emiliana huxleyi*” dominates organism-level nodes while archaea (9 papers, 1.8% of the corpus) and viruses (10 papers, 2.0%) are virtually absent (Section 3.7) — underscores the urgency of extending these molecular approaches to under-studied lineages. Comparative multi-omics across different evolutionary lineages, leveraging reference gene catalogs and biome-specific genome sets, will reveal fundamental principles governing acidification responses and identify universal versus lineage-specific adaptation mechanisms (Eugene L. Madsen, 2011; Wannicke et al., 2018; Whitehead, 2012).

4.2 Expansion of Temporal and Spatial Research Scales

The geographic imbalances documented in Section 3.5 are compounded by a temporal limitation: the keyword co-occurrence network (Section 3.7) shows that geographically specific terms emerged only in the later phases of the field's development — “Mediterranean Sea” first appeared as a keyword node in 2017, while “Yellow Sea” and “East China Sea” emerged as late as 2021, suggesting that most existing studies are not yet spatially resolved at the regional-sea level. Concurrently, the keyword timeline (Section 3.9) reveals that process-oriented terms such as “dynamics” and “food web” appeared only after 2016, indicating a recent but still incomplete transition from static snapshots toward understanding temporal trajectories of microbial community change under sustained acidification. These patterns jointly indicate that the field needs both broader geographic coverage and longer observational time horizons. The pathway forward is clear: couple long-term OA chemistry with microbial ecology at basin scales. The Global Ocean Acidification Observing Network provides the coordination framework, while Biogeochemical-Argo offers vertically resolved, year-round pH and biogeochemical profiles in regions previously under-sampled (Claustre et al., 2020; Tilbrook et al., 2019). Time-series programs demonstrate how sustained observations resolve secular pH decline and variability—an evidentiary base to benchmark models of microbial dynamics under OA (Harper et al., 2023).

Future research initiatives should establish global observatory networks leveraging autonomous underwater vehicles equipped with advanced chemical and biological sensors, enabling continuous monitoring of microbial community structure and function across diverse marine environments (Benway et al., 2019). To harmonize biology across observatories, standardized microbial protocols from global campaigns should be embedded in future OA observatories and mesocosm programs to ensure cross-basin comparability of community structure and function (Kopf et al., 2015). The development of standardized protocols for long-term microbial community monitoring will enable comparative studies across different ocean basins, revealing how regional environmental variability influences acidification impacts (Brown et al., 2018; Godbold & Solan, 2013). Future research will integrate physical oceanography, biogeochemistry, and microbial ecology measurements to provide comprehensive understanding of how acidification effects cascade through marine ecosystems over decadal timescales (Leray et al., 2021; Sutton et al., 2022).

4.3 Multiple Stressor Interaction and Climate Change Integration

In the keyword co-occurrence network (Section 3.7), “temperature” occupies an intermediate bridging position between the core OA cluster and biological response clusters, and the cluster analysis (Section 3.8) identified “interactive effect” as a significant term in Cluster #3 (score = 8.52). However, the manual content analysis of the “responses” burst (Section 3.10) shows that only 47% of recent papers explicitly adopt multi-stressor designs, and no independent multi-

stressor keyword has achieved burst strength comparable to single-factor terms. This gap between the recognized importance of multi-stressor interactions and their relatively modest bibliometric footprint motivates the following direction. Experimental evidence from *Vibrio harveyi* adaptation studies across a pH range mimicking pre-industrial (8.5), present (8.0), and projected future (7.0–7.5) ocean conditions at multiple temperatures (20–30 °C) demonstrates that temperature and pH interact non-additively: elevated temperature, rather than pH alone, was the primary driver of reduced cell culturability at 30 °C, whereas acidification exerted a stronger independent effect at 25 °C (K. Gundogdu et al., 2023). Such non-linear temperature–pH interactions highlight the inadequacy of single-factor OA experimental designs. Next-generation facilities must implement orthogonal control of pCO₂, temperature, O₂, and nutrients, enabling detection of non-linear responses that are obscured in single-factor experiments. Meanwhile, progress across model hierarchies — from trait-based ecosystem frameworks to Earth system models — should explicitly account for microbially driven redox processes and evolutionary adaptations under combined stressors, to better connect lab-scale findings with Earth-scale predictions (Boyd et al., 2018a).

These multiple stressor approaches will be particularly critical for understanding how acidification impacts cascade through marine food webs under realistic environmental conditions (Hu et al., 2024). The development of Earth system models that integrate microbial community dynamics with global biogeochemical cycles represents a major frontier for understanding how local acidification impacts scale to global carbon cycling patterns (Subramaniam et al., 2016; Tagliabue et al., 2011).

4.4 Next-Generation Experimental Systems, Computational Integration, and Science-Policy Translation

The disconnect between organism-level keyword clusters (#0 “growth,” #1 “elemental ratios,” #4 “reduced calcification”) and ecosystem-scale clusters (#3 “climate change,” #6 “carbon”) in the clustering map (Section 3.8) reveals a persistent scale-integration gap: individual-level mechanistic knowledge is not yet being systematically connected to ecosystem-level biogeochemical predictions. Bridging this gap requires advances in experimental infrastructure, computational frameworks, and science-policy. Extending this trajectory, in situ manipulation platforms such as Free-Ocean CO₂ Enrichment enable long-duration, ecologically realistic OA exposures that preserve natural food-web interactions (Gattuso et al., 2014). Embedding ecogenomic sensors and lab-on-chip pH/chemistry analyzers within mesocosms and benthic arrays will permit real-time tracking of gene expression, metabolites, and carbonate chemistry at the exposure scale organisms actually experience.

Advanced bioreactor systems will enable precise control of multiple environmental variables while maintaining realistic ecological complexity (Hu et al., 2021). These systems will incorporate microfluidic technologies for studying microbial interactions at microscales,

automated sampling systems for high-temporal resolution measurements, and in situ chemical sensors for continuous monitoring of carbonate system parameters. At micro-scales, microfluidics can recreate diffusive gradients and patch dynamics to mechanistically link cell-level behavior to population outcomes under acidification (Depetris et al., 2022; Ugolini et al., 2024).

The development of portable, field-deployable experimental systems will enable acidification research in previously inaccessible environments, from deep-sea hydrothermal vents to polar ice-edge ecosystems (Hakonen et al., 2013; Sastri et al., 2019). These technological advances will expand the geographic and environmental scope of acidification research, revealing how microbial communities respond to acidification across the full spectrum of marine environments (Treusch et al., 2009).

Equally important is the computational infrastructure needed to synthesize data across these experimental scales. Community platforms such as MGnify and KBase already provide harmonized multi-omics analyses, while machine learning approaches applied to integrated datasets enable predictive models of microbial community responses under projected acidification scenarios (Dutta et al., 2022; Rubbens et al., 2023). Digital twins of the ocean will offer high-resolution simulation environments where OA-relevant microbial processes can be coupled to circulation, biogeochemistry, and management scenarios (Canonica et al., 2019).

These experimental and computational advances will also strengthen the policy-science interface. The keyword co-occurrence network shows that “climate change” (Cluster #3) and “carbon” (Cluster #6) are among the largest thematic clusters yet remain poorly connected to organism-level clusters, suggesting that translating microbial-scale findings into ecosystem-level policy indicators remains an unresolved challenge. Embedding OA-microbiology outputs in Global Ocean Observing System frameworks and digital-twin dashboards will translate this complexity into decision-ready information for conservation planning and climate mitigation (Miloslavich et al., 2018).

5. Limitations and Methodological Considerations

Several methodological limitations should be considered when interpreting results from this bibliometric analysis. The exclusive reliance on Web of Science data may introduce coverage bias, potentially underrepresenting research published in journals not indexed by this database or publications in languages other than English. The search strategy, while designed to optimize precision and recall, may have excluded some relevant publications that employ different terminological conventions or focus on related but distinct research questions.

Citation-based measures inherently reflect past research impact rather than current scientific quality or future potential, potentially creating temporal biases that favor older publications over recent contributions. Self-citation patterns and disciplinary citation practices may

influence network structures in ways that do not accurately reflect knowledge flows or collaborative relationships. The 20-year temporal scope, while comprehensive for this relatively young field, may not capture important earlier foundational work that established the conceptual framework for OA research.

These limitations are inherent to bibliometric approaches and do not invalidate the principal findings, but rather provide important context for interpreting results and identifying opportunities for complementary analytical approaches. The comprehensive multi-method framework employed in this study helps mitigate individual methodological limitations by providing multiple perspectives on the same underlying research landscape, enhancing the robustness and reliability of principal conclusions.

Finally, while structural variation analysis and co-citation clustering provide valuable insights into the intellectual organization of the field, they do not directly evaluate the quality, reproducibility, or methodological rigor of the underlying studies. Integrating bibliometric findings with systematic reviews or meta-analyses could offer a more comprehensive assessment of research progress.

6. Conclusion

This bibliometric analysis of 495 publications (2005-2025) provides the first microbiology-specific quantitative mapping of the OA research landscape, addressing two research questions.

Regarding RQ1 (evolution, collaboration, and structural features): the field exhibits a three-phase trajectory of emergence (2005-2010), rapid growth peaking in 2017 followed by fluctuation (2011-2021), and post-peak adjustment (2022-2025), though the most recent counts may be affected by indexing lag. The global collaboration network spans 53 countries with a triadic production leadership of China (143 papers), Germany (123), and the USA (116), while France (centrality = 0.72) and England (centrality = 0.47) serve as the primary international bridges. GEOMAR functions as the central institutional hub, largely attributable to its coordination of large-scale mesocosm programs. However, geographic participation remains unbalanced, with Africa (6 papers), Southeast Asia (7 papers), and Small Island Developing States (absent) severely underrepresented, a pattern that reflects both capacity constraints and systemic knowledge inequities amplified by the English-language bias of the Web of Science database.

Regarding RQ2 (thematic structure, frontiers, and blind spots): keyword and co-citation analyses reveal a clear paradigm shift from foundational carbonate chemistry (Cluster #8) toward community-level and functional ecological investigations (Clusters #0, #3, #5). Manual content analysis of the dominant burst term “responses” identifies three specific sub-frontiers: physiological-metabolic adjustments (68% of burst papers), community-structural shifts (50%), and multi-stressor interaction effects (47%), with evolutionary and epigenetic responses (6%)

representing a largely unexplored frontier. Taxonomically, the field remains biased toward coccolithophores, with archaea (9 papers, 1.8%) and viruses (10 papers, 2.0%) conspicuously underrepresented despite their ecological significance, and neither group registering as a node in the 217-keyword co-occurrence network. The four evidence-linked research directions proposed in Section 4 are each anchored to specific bibliometric gaps identified herein and offer a data-driven framework for prioritizing next-generation OA-microbe research.

Interest statement: On behalf of all authors, the corresponding author states that there is no conflict of interest.

Declaration of generative AI and AI-assisted technologies in the manuscript preparation process: During the preparation of this work the authors used ChatGPT 5.5 in order to improve readability and language. After using this tool, the authors reviewed and edited the content as needed and take full responsibility for the content of the publication.

References

- Aberle, N., Schulz, K. G., Stuhr, A., Malzahn, A. M., Ludwig, A., & Riebesell, U. (2013). High tolerance of microzooplankton to ocean acidification in an Arctic coastal plankton community. *Biogeosciences*, 10(3), 1471-1481. <https://doi.org/10.5194/bg-10-1471-2013>
- Abreu, A., Bourgois, E., Gristwood, A., Troublé, R., Acinas, S. G., Bork, P., Boss, E., Bowler, C., Budinich, M., Chaffron, S., de Vargas, C., Delmont, T. O., Eveillard, D., Guidi, L., Iudicone, D., Kandels, S., Morlon, H., Lombard, F., Pepperkok, R.,...European Marine Biological Resource Centre - European Research Infrastructure, C. (2022). Priorities for ocean microbiome research. *Nature Microbiology*, 7(7), 937-947. <https://doi.org/10.1038/s41564-022-01145-5>
- Aguayo, P., Campos, V. L., Henríquez, C., Olivares, F., De la Iglesia, R., Ulloa, O., & Vargas, C. A. (2020). The Influence of pCO₂-Driven Ocean Acidification on Open Ocean Bacterial Communities during A Short-Term Microcosm Experiment in the Eastern Tropical South Pacific (ETSP) off Northern Chile. *Microorganisms*, 8(12). <https://doi.org/10.3390/microorganisms8121924>
- Aguayo, P., Campos, V. L., Henríquez, C., Olivares, F., Iglesia, R. D., Ulloa, O., & Vargas, C. A. (2020). The Influence of pCO₂-Driven Ocean Acidification on Open Ocean Bacterial Communities during A Short-Term Microcosm Experiment in the Eastern Tropical South Pacific (ETSP) off Northern Chile. *Microorganisms*, 8(12), Article 1924. <https://doi.org/10.3390/microorganisms8121924>
- Ahme, A., Von Jackowski, A., McPherson, R. A., Wolf, K. K. E., Hoppmann, M., Neuhaus, S., & John, U. (2023). Winners and Losers of Atlantification: The Degree of Ocean Warming Affects the Structure of Arctic Microbial Communities. *Genes*, 14(3). <https://doi.org/10.3390/genes14030623>
- Albright, R., Caldeira, L., Hosfelt, J., Kwiatkowski, L., Maclaren, J. K., Mason, B. M., Nebuchina, Y., Ninokawa, A., Pongratz, J., Ricke, K. L., Rivlin, T., Schneider, K., Sesboüé, M., Shamberger, K., Silverman, J., Wolfe, K., Zhu, K., & Caldeira, K. (2016). Reversal of ocean acidification enhances net coral reef calcification. *Nature*, 531(7594), 362-365.

- 1106 <https://doi.org/10.1038/nature17155>
- 1107 Anil, A. C., Bach, L. T., Taucher, J., Boxhammer, T., Ludwig, A., Achterberg, E. P., Algueró-
- 1108 Muñiz, M., Anderson, L. G., Bellworthy, J., Büdenbender, J., Czerny, J., Ericson, Y., Esposito,
- 1109 M., Fischer, M., Haunost, M., Hellemann, D., Horn, H. G., Hornick, T., Meyer, J.,...Riebesell,
- 1110 U. (2016). Influence of Ocean Acidification on a Natural Winter-to-Summer Plankton
- 1111 Succession: First Insights from a Long-Term Mesocosm Study Draw Attention to Periods of
- 1112 Low Nutrient Concentrations. *Plos One*, 11(8). <https://doi.org/10.1371/journal.pone.0159068>
- 1113 Azam, F., & Malfatti, F. (2007). Microbial structuring of marine ecosystems. *Nature Reviews*
- 1114 *Microbiology*, 5(10), 782-791. <https://doi.org/10.1038/nrmicro1747>
- 1115 Bach, L. T., Riebesell, U., & Schulz, K. G. (2011). Distinguishing between the effects of ocean
- 1116 acidification and ocean carbonation in the coccolithophore *Emiliana huxleyi*. *Limnology and*
- 1117 *Oceanography*, 56(6), 2040-2050. <https://doi.org/10.4319/lo.2011.56.6.2040>
- 1118 Barreto, M. M., Ziegler, M., Venn, A., Tambutté, E., Zoccola, D., Tambutté, S., Allemand, D.,
- 1119 Antony, C. P., Voolstra, C. R., & Aranda, M. (2021). Effects of Ocean Acidification on Resident
- 1120 and Active Microbial Communities of *Stylophora pistillata*. *Frontiers in Microbiology*, 12.
- 1121 <https://doi.org/10.3389/fmicb.2021.707674>
- 1122 Barrett, R. C., Carter, B. R., Fassbender, A. J., Tilbrook, B., Woosley, R. J., Azetsu-Scott, K.,
- 1123 Feely, R. A., Goyet, C., Ishii, M., Murata, A., & Pérez, F. F. (2025). Biological Responses to
- 1124 Ocean Acidification Are Changing the Global Ocean Carbon Cycle. *Global Biogeochemical*
- 1125 *Cycles*, 39(3). <https://doi.org/10.1029/2024gb008358>
- 1126 Beaufort, L., Probert, I., de Garidel-Thoron, T., Bendif, E. M., Ruiz-Pino, D., Metzl, N., Goyet,
- 1127 C., Buchet, N., Coupel, P., Grelaud, M., Rost, B., Rickaby, R. E. M., & de Vargas, C. (2011).
- 1128 Sensitivity of coccolithophores to carbonate chemistry and ocean acidification. *Nature*,
- 1129 476(7358), 80-83. <https://doi.org/10.1038/nature10295>
- 1130 Benway, H. M., Lorenzoni, L., White, A. E., Fiedler, B., Levine, N. M., Nicholson, D. P.,
- 1131 DeGrandpre, M. D., Sosik, H. M., Church, M. J., O'Brien, T. D., Leinen, M., Weller, R. A.,
- 1132 Karl, D. M., Henson, S. A., & Letelier, R. M. (2019). Ocean Time Series Observations of
- 1133 Changing Marine Ecosystems: An Era of Integration, Synthesis, and Societal Applications.
- 1134 *Frontiers in Marine Science*, 6. <https://doi.org/10.3389/fmars.2019.00393>
- 1135 Bermúdez, J. R., Riebesell, U., Larsen, A., & Winder, M. (2016). Ocean acidification reduces
- 1136 transfer of essential biomolecules in a natural plankton community. *Scientific Reports*, 6(1),
- 1137 27749. <https://doi.org/10.1038/srep27749>
- 1138 Boyd, P. W., Collins, S., Dupont, S., Fabricius, K., Gattuso, J.-P., Havenhand, J., Hutchins, D.
- 1139 A., Riebesell, U., Rintoul, M. S., Vichi, M., Biswas, H., Ciotti, A., Gao, K., Gehlen, M., Hurd,
- 1140 C. L., Kurihara, H., McGraw, C. M., Navarro, J. M., Nilsson, G. E.,...Pörtner, H.-O. (2018a).
- 1141 Experimental strategies to assess the biological ramifications of multiple drivers of global ocean
- 1142 change—A review. *Global Change Biology*, 24(6), 2239-2261.
- 1143 <https://doi.org/https://doi.org/10.1111/gcb.14102>
- 1144 Boyd, P. W., Collins, S., Dupont, S., Fabricius, K., Gattuso, J. P., Havenhand, J., Hutchins, D.
- 1145 A., Riebesell, U., Rintoul, M. S., Vichi, M., Biswas, H., Ciotti, A., Gao, K., Gehlen, M., Hurd,
- 1146 C. L., Kurihara, H., McGraw, C. M., Navarro, J. M., Nilsson, G. E.,...Pörtner, H. O. (2018b).
- 1147 Experimental strategies to assess the biological ramifications of multiple drivers of global ocean
- 1148 change—A review. *Global Change Biology*, 24(6), 2239-2261.
- 1149 <https://doi.org/10.1111/gcb.14102>

- Brandenburg, K. M., Velthuis, M., & Van de Waal, D. B. (2019). Meta-analysis reveals enhanced growth of marine harmful algae from temperate regions with warming and elevated CO₂ levels. *Global Change Biology*, 25(8), 2607-2618. <https://doi.org/10.1111/gcb.14678>
- Breitbarth, E., Bellerby, R. J., Neill, C. C., Ardelan, M. V., Meyerhöfer, M., Zöllner, E., Croot, P. L., & Riebesell, U. (2010). Ocean acidification affects iron speciation during a coastal seawater mesocosm experiment. *Biogeosciences*, 7(3), 1065-1073. <https://doi.org/10.5194/bg-7-1065-2010>
- Brown, M. V., van de Kamp, J., Ostrowski, M., Seymour, J. R., Ingleton, T., Messer, L. F., Jeffries, T., Siboni, N., Laverock, B., Bibiloni-Isaksson, J., Nelson, T. M., Coman, F., Davies, C. H., Frampton, D., Rayner, M., Goossen, K., Robert, S., Holmes, B., Abell, G. C. J.,...Bodrossy, L. (2018). Systematic, continental scale temporal monitoring of marine pelagic microbiota by the Australian Marine Microbial Biodiversity Initiative. *Scientific Data*, 5(1). <https://doi.org/10.1038/sdata.2018.130>
- Brussaard, C. P. D., Noordeloos, A. A. M., Witte, H., Collenteur, M. C. J., Schulz, K., Ludwig, A., & Riebesell, U. (2013). Arctic microbial community dynamics influenced by elevated CO₂ levels. *Biogeosciences*, 10(2), 719-731. <https://doi.org/10.5194/bg-10-719-2013>
- Campen, H. I., Arévalo-Martínez, D. L., Artioli, Y., Brown, I. J., Kitidis, V., Lessin, G., Rees, A. P., & Bange, H. W. (2021). The role of a changing Arctic Ocean and climate for the biogeochemical cycling of dimethyl sulphide and carbon monoxide. *Ambio*, 51(2), 411-422. <https://doi.org/10.1007/s13280-021-01612-z>
- Canonico, G., Buttigieg, P. L., Montes, E., Muller-Karger, F. E., Stepien, C., Wright, D., Benson, A., Helmuth, B., Costello, M., Sousa-Pinto, I., Saeedi, H., Newton, J., Appeltans, W., Bednaršek, N., Bodrossy, L., Best, B. D., Brandt, A., Goodwin, K. D., Iken, K.,...Murton, B. (2019). Global Observational Needs and Resources for Marine Biodiversity. *Frontiers in Marine Science*, 6. <https://doi.org/10.3389/fmars.2019.00367>
- Carter, B. R., Sharp, J. D., Dickson, A. G., Álvarez, M., Fong, M. B., García-Ibáñez, M. I., Woosley, R. J., Takeshita, Y., Barbero, L., Byrne, R. H., Cai, W. J., Chierici, M., Clegg, S. L., Easley, R. A., Fassbender, A. J., Flegler, K. L., Li, X., Martín-Mayor, M., Schockman, K. M., & Wang, Z. A. (2023). Uncertainty sources for measurable ocean carbonate chemistry variables. *Limnology and Oceanography*, 69(1), 1-21. <https://doi.org/10.1002/lno.12477>
- Chakraborty, S., Paidi, M. K., Dhinakaran, I., Sivakumar, M., Clements, C., Thirumurugan, N. K., & Sivakumar, L. (2024a). Adaptive mechanism of the marine bacterium *Pseudomonas sihuiensis*-BFB-6S towards pCO₂ variation: Insights into synthesis of extracellular polymeric substances and physiochemical modulation. *International Journal of Biological Macromolecules*, 261. <https://doi.org/10.1016/j.ijbiomac.2024.129860>
- Chakraborty, S., Paidi, M. K., Dhinakaran, I., Sivakumar, M., Clements, C., Thirumurugan, N. K., & Sivakumar, L. (2024b). Adaptive mechanism of the marine bacterium *Pseudomonas sihuiensis*-BFB-6S towards pCO₂ variation: Insights into synthesis of extracellular polymeric substances and physiochemical modulation. *International Journal of Biological Macromolecules*, 261, 129860. <https://doi.org/10.1016/j.ijbiomac.2024.129860>
- Chen, C. (2004). Searching for intellectual turning points: Progressive knowledge domain visualization. *Proceedings of the National Academy of Sciences*, 101(suppl_1), 5303-5310. <https://doi.org/10.1073/pnas.0307513100>

- Chen, C. (2005). CiteSpace II: Detecting and visualizing emerging trends and transient patterns in scientific literature. *Journal of the American Society for Information Science and Technology*, 57(3), 359-377. <https://doi.org/10.1002/asi.20317>
- Chen, C. (2017). Science Mapping: A Systematic Review of the Literature. *Journal of Data and Information Science*, 2(2), 1-40. <https://doi.org/10.1515/jdis-2017-0006>
- Chin, W.-C., Kübler, J. E., & Dudgeon, S. R. (2015). Predicting Effects of Ocean Acidification and Warming on Algae Lacking Carbon Concentrating Mechanisms. *Plos One*, 10(7). <https://doi.org/10.1371/journal.pone.0132806>
- Claustre, H., Johnson, K. S., & Takeshita, Y. (2020). Observing the Global Ocean with Biogeochemical-Argo. *Annual Review of Marine Science*, 12(Volume 12, 2020), 23-48. <https://doi.org/https://doi.org/10.1146/annurev-marine-010419-010956>
- Cohen, A. L., McCorkle, D. C., de Putron, S., Gaetani, G. A., & Rose, K. A. (2009). Morphological and compositional changes in the skeletons of new coral recruits reared in acidified seawater: Insights into the biomineralization response to ocean acidification. *Geochemistry, Geophysics, Geosystems*, 10(7). <https://doi.org/10.1029/2009gc002411>
- Cornwall, C. E., Boyd, P. W., McGraw, C. M., Hepburn, C. D., Pilditch, C. A., Morris, J. N., Smith, A. M., & Hurd, C. L. (2014). Diffusion Boundary Layers Ameliorate the Negative Effects of Ocean Acidification on the Temperate Coralline Macroalga *Arthrocardia corymbosa*. *PLOS ONE*, 9(5), e97235. <https://doi.org/10.1371/journal.pone.0097235>
- Cornwall, C. E., & Hurd, C. L. (2016). Experimental design in ocean acidification research: problems and solutions. *ICES Journal of Marine Science*, 73(3), 572-581. <https://doi.org/10.1093/icesjms/fsv118>
- Crump, B. C., & Bowen, J. L. (2024). The Microbial Ecology of Estuarine Ecosystems. *Annual Review of Marine Science*, 16(Volume 16, 2024), 335-360. <https://doi.org/https://doi.org/10.1146/annurev-marine-022123-101845>
- Currie, A. R., Tait, K., Parry, H., de Francisco-Mora, B., Hicks, N., Osborn, A. M., Widdicombe, S., & Stahl, H. (2017). Marine Microbial Gene Abundance and Community Composition in Response to Ocean Acidification and Elevated Temperature in Two Contrasting Coastal Marine Sediments. *Frontiers in Microbiology*, 8. <https://doi.org/10.3389/fmicb.2017.01599>
- D'Amario, B., Pérez, C., Grelaud, M., Pitta, P., Krasakopoulou, E., & Ziveri, P. (2020). Coccolithophore community response to ocean acidification and warming in the Eastern Mediterranean Sea: results from a mesocosm experiment. *Scientific Reports*, 10(1), 12637. <https://doi.org/10.1038/s41598-020-69519-5>
- Dam, H. G., Bach, L. T., Alvarez-Fernandez, S., Hornick, T., Stühr, A., & Riebesell, U. (2017). Simulated ocean acidification reveals winners and losers in coastal phytoplankton. *Plos One*, 12(11). <https://doi.org/10.1371/journal.pone.0188198>
- Depetris, A., Tagliavini, G., Peter, H., Köhl, M., Holzner, M., & Battin, T. J. (2022). Biophysical properties at patch scale shape the metabolism of biofilm landscapes. *npj Biofilms and Microbiomes*, 8(1), 5. <https://doi.org/10.1038/s41522-022-00269-0>
- Deppeler, S., Petrou, K., Schulz, K. G., Westwood, K., Pearce, I., McKinlay, J., & Davidson, A. (2018). Ocean acidification of a coastal Antarctic marine microbial community reveals a critical threshold for CO₂ tolerance in phytoplankton productivity. *Biogeosciences*, 15(1), 209-231. <https://doi.org/10.5194/bg-15-209-2018>
- Doney, S. C., Balch, W. M., Fabry, V. J., & Feely, R. A. (2009). OCEAN ACIDIFICATION A

- 1238 CRITICAL EMERGING PROBLEM FOR THE OCEAN SCIENCES. *Oceanography*, 22(4),
 1239 16-25. <http://www.jstor.org/stable/24861020>
- 1240 Doney, S. C., Busch, D. S., Cooley, S. R., & Kroeker, K. J. (2020). The Impacts of Ocean
 1241 Acidification on Marine Ecosystems and Reliant Human Communities. *Annual Review of*
 1242 *Environment and Resources*, 45(Volume 45, 2020), 83-112.
 1243 <https://doi.org/https://doi.org/10.1146/annurev-environ-012320-083019>
- 1244 Doney, S. C., Fabry, V. J., Feely, R. A., & Kleypas, J. A. (2009a). Ocean Acidification: The
 1245 Other CO₂ Problem. *Annual Review of Marine Science*, 1(Volume 1, 2009), 169-192.
 1246 <https://doi.org/https://doi.org/10.1146/annurev.marine.010908.163834>
- 1247 Doney, S. C., Fabry, V. J., Feely, R. A., & Kleypas, J. A. (2009b). Ocean Acidification: The
 1248 Other CO₂ Problem. *Annual Review of Marine Science*, 1(1), 169-192.
 1249 <https://doi.org/10.1146/annurev.marine.010908.163834>
- 1250 Dore, J. E., Lukas, R., Sadler, D. W., Church, M. J., & Karl, D. M. (2009). Physical and
 1251 biogeochemical modulation of ocean acidification in the central North Pacific. *Proceedings of*
 1252 *the National Academy of Sciences*, 106(30), 12235-12240.
 1253 <https://doi.org/10.1073/pnas.0906044106>
- 1254 Dupont, S., Rokitta, S. D., John, U., & Rost, B. (2012). Ocean Acidification Affects Redox-
 1255 Balance and Ion-Homeostasis in the Life-Cycle Stages of *Emiliana huxleyi*. *Plos One*, 7(12).
 1256 <https://doi.org/10.1371/journal.pone.0052212>
- 1257 Dutta, A., Goldman, T., Keating, J., Burke, E., Williamson, N., Dirmeier, R., Bowman, J. S., &
 1258 Denef, V. J. (2022). Machine Learning Predicts Biogeochemistry from Microbial Community
 1259 Structure in a Complex Model System. *Microbiology Spectrum*, 10(1).
 1260 <https://doi.org/10.1128/spectrum.01909-21>
- 1261 Endres, S., Galgani, L., Riebesell, U., Schulz, K.-G., & Engel, A. (2014). Stimulated Bacterial
 1262 Growth under Elevated pCO₂: Results from an Off-Shore Mesocosm Study. *PLOS ONE*, 9(6),
 1263 e99228. <https://doi.org/10.1371/journal.pone.0099228>
- 1264 Fabry, V. J., Seibel, B. A., Feely, R. A., & Orr, J. C. (2008). Impacts of ocean acidification on
 1265 marine fauna and ecosystem processes. *ICES Journal of Marine Science*, 65(3), 414-432.
 1266 <https://doi.org/10.1093/icesjms/fsn048>
- 1267 Feely, R. A., Sabine, C. L., Lee, K., Berelson, W., Kleypas, J., Fabry, V. J., & Millero, F. J.
 1268 (2004). Impact of Anthropogenic CO₂ on the CaCO₃ System in the
 1269 Oceans. *Science*, 305(5682), 362-366. <https://doi.org/doi:10.1126/science.1097329>
- 1270 Fennel, K., Mattern, J. P., Doney, S. C., Bopp, L., Moore, A. M., Wang, B., & Yu, L. (2022).
 1271 Ocean biogeochemical modelling. *Nature Reviews Methods Primers*, 2(1).
 1272 <https://doi.org/10.1038/s43586-022-00154-2>
- 1273 Ferderer, A., Schulz, K. G., Riebesell, U., Baker, K. G., Chase, Z., & Bach, L. T. (2024).
 1274 Investigating the effect of silicate- and calcium-based ocean alkalinity enhancement on diatom
 1275 silicification. *Biogeosciences*, 21(11), 2777-2794. <https://doi.org/10.5194/bg-21-2777-2024>
- 1276 Flynn, K. J., Blackford, J. C., Baird, M. E., Raven, J. A., Clark, D. R., Beardall, J., Brownlee,
 1277 C., Fabian, H., & Wheeler, G. L. (2012). Changes in pH at the exterior surface of plankton with
 1278 ocean acidification. *Nature Climate Change*, 2(7), 510-513.
 1279 <https://doi.org/10.1038/nclimate1489>
- 1280 Gao, K., Beardall, J., Häder, D.-P., Hall-Spencer, J. M., Gao, G., & Hutchins, D. A. (2019a).
 1281 Effects of Ocean Acidification on Marine Photosynthetic Organisms Under the Concurrent

- 1282 Influences of Warming, UV Radiation, and Deoxygenation [Review]. *Frontiers in Marine*
 1283 *Science, Volume 6 - 2019*. <https://doi.org/10.3389/fmars.2019.00322>
- 1284 Gao, K., Beardall, J., Häder, D.-P., Hall-Spencer, J. M., Gao, G., & Hutchins, D. A. (2019b).
 1285 Effects of Ocean Acidification on Marine Photosynthetic Organisms Under the Concurrent
 1286 Influences of Warming, UV Radiation, and Deoxygenation. *Frontiers in Marine Science*, 6.
 1287 <https://doi.org/10.3389/fmars.2019.00322>
- 1288 Gao, K., & Campbell, D. A. (2014). Photophysiological responses of marine diatoms to
 1289 elevated CO₂ and decreased pH: a review. *Functional Plant Biology*, 41(5).
 1290 <https://doi.org/10.1071/fp13247>
- 1291 Gao, K., Gao, G., Wang, Y., & Dupont, S. (2020). Impacts of ocean acidification under multiple
 1292 stressors on typical organisms and ecological processes. *Marine Life Science & Technology*,
 1293 2(3), 279-291. <https://doi.org/10.1007/s42995-020-00048-w>
- 1294 Gao, K., Helbling, E. W., Häder, D. P., & Hutchins, D. A. (2012). Responses of marine primary
 1295 producers to interactions between ocean acidification, solar radiation, and warming. *Marine*
 1296 *Ecology Progress Series*, 470, 167-189. <https://doi.org/10.3354/meps10043>
- 1297 Gao, K., Xu, J., Gao, G., Li, Y., Hutchins, D. A., Huang, B., Wang, L., Zheng, Y., Jin, P., Cai,
 1298 X., Häder, D.-P., Li, W., Xu, K., Liu, N., & Riebesell, U. (2012). Rising CO₂ and increased
 1299 light exposure synergistically reduce marine primary productivity. *Nature Climate Change*, 2(7),
 1300 519-523. <https://doi.org/10.1038/nclimate1507>
- 1301 Gattuso, J. P., Kirkwood, W., Barry, J. P., Cox, E., Gazeau, F., Hansson, L., Hendriks, I., Kline,
 1302 D. I., Mahacek, P., Martin, S., McElhany, P., Peltzer, E. T., Reeve, J., Roberts, D., Saderne, V.,
 1303 Tait, K., Widdicombe, S., & Brewer, P. G. (2014). Free-ocean CO₂ enrichment
 1304 (FOCE) systems: present status and future developments. *Biogeosciences*, 11(15), 4057-4075.
 1305 <https://doi.org/10.5194/bg-11-4057-2014>
- 1306 Geng, Y., Zhang, X., Gao, J., Yan, Y., & Chen, L. (2024). Bibliometric analysis of sustainable
 1307 tourism using CiteSpace. *Technological Forecasting and Social Change*, 202.
 1308 <https://doi.org/10.3390/w15234162>
- 1309 Geng, Y. Q., Zhang, X. R., Gao, J., Yan, Y., & Chen, L. Y. (2024). Bibliometric analysis of
 1310 sustainable tourism using CiteSpace. *Technological Forecasting and Social Change*, 202,
 1311 Article 123310. <https://doi.org/10.1016/j.techfore.2024.123310>
- 1312 Glibert, P. M., & Mitra, A. (2022). From webs, loops, shunts, and pumps to microbial
 1313 multitasking: Evolving concepts of marine microbial ecology, the mixoplankton paradigm, and
 1314 implications for a future ocean. *Limnology and Oceanography*, 67(3), 585-597.
 1315 <https://doi.org/https://doi.org/10.1002/lno.12018>
- 1316 Godbold, J. A., & Solan, M. (2013). Long-term effects of warming and ocean acidification are
 1317 modified by seasonal variation in species responses and environmental conditions.
 1318 *Philosophical Transactions of the Royal Society B: Biological Sciences*, 368(1627).
 1319 <https://doi.org/10.1098/rstb.2013.0186>
- 1320 Goldenberg, S. U., Nagelkerken, I., Ferreira, C. M., Ullah, H., & Connell, S. D. (2017). Boosted
 1321 food web productivity through ocean acidification collapses under warming. *Global Change*
 1322 *Biology*, 23(10), 4177-4184. <https://doi.org/10.1111/gcb.13699>
- 1323 Griffen, B. D., Belgrad, B. A., Cannizzo, Z. J., Knotts, E. R., & Hancock, E. R. (2016).
 1324 Rethinking our approach to multiple stressor studies in marine environments. *Marine Ecology*
 1325 *Progress Series*, 543, 273-281. <https://doi.org/10.3354/meps11595>

- Grosse, J., Endres, S., & Engel, A. (2020). Ocean acidification modifies biomolecule composition in organic matter through complex interactions. *Scientific Reports*, 10(1), 20599. <https://doi.org/10.1038/s41598-020-77645-3>
- Gunderson, A. R., Armstrong, E. J., & Stillman, J. H. (2016). Multiple Stressors in a Changing World: The Need for an Improved Perspective on Physiological Responses to the Dynamic Marine Environment. *Annual Review of Marine Science*, 8(1), 357-378. <https://doi.org/10.1146/annurev-marine-122414-033953>
- Gundogdu, K., Iturriza, A. O., Orruño, M., Montánchez, I., Eguiraun, H., Martinez, I., Arana, I., & Kaberdin, V. R. (2023). Addressing the Joint Impact of Temperature and pH on *Vibrio harveyi* Adaptation in the Time of Climate Change. *Microorganisms*, 11(4), Article 1075. <https://doi.org/10.3390/microorganisms11041075>
- Gundogdu, K., Orus Iturriza, A., Orruño, M., Montánchez, I., Eguiraun, H., Martinez, I., Arana, I., & Kaberdin, V. R. (2023). Addressing the Joint Impact of Temperature and pH on *Vibrio harveyi* Adaptation in the Time of Climate Change. *Microorganisms*, 11(4). <https://doi.org/10.3390/microorganisms11041075>
- Hakonen, A., Anderson, L. G., Engelbrektsson, J., Hulth, S., & Karlson, B. (2013). A potential tool for high-resolution monitoring of ocean acidification. *Analytica Chimica Acta*, 786, 1-7. <https://doi.org/10.1016/j.aca.2013.04.040>
- Hancock, A. M., King, C. K., Stark, J. S., McMinn, A., & Davidson, A. T. (2020). Effects of ocean acidification on Antarctic marine organisms: A meta-analysis. *Ecology and Evolution*, 10(10), 4495-4514. <https://doi.org/10.1002/ece3.6205>
- Harper, K. L., Lamarche, C., Hartley, A., Peylin, P., Ottlé, C., Bastrikov, V., San Martín, R., Bohnenstengel, S. I., Kirches, G., Boettcher, M., Shevchuk, R., Brockmann, C., & Defourny, P. (2023). A 29-year time series of annual 300 m resolution plant-functional-type maps for climate models. *Earth Syst. Sci. Data*, 15(3), 1465-1499. <https://doi.org/10.5194/essd-15-1465-2023>
- Harvey, B. P., Gwynn-Jones, D., & Moore, P. J. (2013). Meta-analysis reveals complex marine biological responses to the interactive effects of ocean acidification and warming. *Ecology and Evolution*, 3(4), 1016-1030. <https://doi.org/https://doi.org/10.1002/ece3.516>
- Hassenrück, C., Fink, A., Lichtschlag, A., Tegetmeyer, H. E., de Beer, D., Ramette, A., & King, G. (2016). Quantification of the effects of ocean acidification on sediment microbial communities in the environment: the importance of ecosystem approaches. *FEMS Microbiology Ecology*, 92(5). <https://doi.org/10.1093/femsec/fiw027>
- Heinze, C., Meyer, S., Goris, N., Anderson, L., Steinfeldt, R., Chang, N., Le Quéré, C., & Bakker, D. C. E. (2015). The ocean carbon sink – impacts, vulnerabilities and challenges. *Earth System Dynamics*, 6(1), 327-358. <https://doi.org/10.5194/esd-6-327-2015>
- Hobday, A. J., Walters, V. M., Stephenson, R. L., Baylis, S., Bessey, C., Boschetti, F., Bulman, C., Contardo, S., Dambacher, J. M., Day, J., Dowling, N. A., Dunstan, P., Eveson, J. P., Farley, J. H., Green, M., Fulton, E. A., Grewe, P., Kunnath, H., Lenton, A.,...Wilcox, C. (2024). Predicting and assessing the impacts of COVID-19 disruption on marine science and sectors in Australia. *Reviews in Fish Biology and Fisheries*, 35(1), 241-278. <https://doi.org/10.1007/s11160-024-09899-3>
- Hofmann, G. E., Barry, J. P., Edmunds, P. J., Gates, R. D., Hutchins, D. A., Klinger, T., & Sewell, M. A. (2010). The Effect of Ocean Acidification on Calcifying Organisms in Marine Ecosystems: An Organism-to-Ecosystem Perspective. *Annual Review of Ecology, Evolution*,

- 1370 *and Systematics*, 41(1), 127-147. <https://doi.org/10.1146/annurev.ecolsys.110308.120227>
- 1371 Holcomb, M., Venn, A. A., Tambutté, E., Tambutté, S., Allemand, D., Trotter, J., & McCulloch,
- 1372 M. (2014). Coral calcifying fluid pH dictates response to ocean acidification. *Scientific Reports*,
- 1373 4(1), 5207. <https://doi.org/10.1038/srep05207>
- 1374 Hu, C., Li, X., He, M., Jiang, P., Long, A., & Xu, J. (2021). Effect of Ocean Acidification on
- 1375 Bacterial Metabolic Activity and Community Composition in Oligotrophic Oceans, Inferred
- 1376 From Short-Term Bioassays. *Frontiers in Microbiology*, 12.
- 1377 <https://doi.org/10.3389/fmicb.2021.583982>
- 1378 Hu, N., Bourdeau, P. E., & Hollander, J. (2024). Responses of marine trophic levels to the
- 1379 combined effects of ocean acidification and warming. *Nature Communications*, 15(1).
- 1380 <https://doi.org/10.1038/s41467-024-47563-3>
- 1381 Hutchins, D. A., Fu, F.-X., Yang, S.-C., John, S. G., Romaniello, S. J., Andrews, M. G., &
- 1382 Walworth, N. G. (2023). Responses of globally important phytoplankton species to olivine
- 1383 dissolution products and implications for carbon dioxide removal via ocean alkalinity
- 1384 enhancement. *Biogeosciences*, 20(22), 4669-4682. <https://doi.org/10.5194/bg-20-4669-2023>
- 1385 Ianora, A., Li, Y., Zhuang, S., Wu, Y., Ren, H., Chen, F., Lin, X., Wang, K., Beardall, J., & Gao,
- 1386 K. (2017). Ocean acidification modulates expression of genes and physiological performance
- 1387 of a marine diatom. *Plos One*, 12(2). <https://doi.org/10.1371/journal.pone.0170970>
- 1388 Iglesias-Rodriguez, M. D., Halloran, P. R., Rickaby, R. E. M., Hall, I. R., Colmenero-Hidalgo,
- 1389 E., Gittins, J. R., Green, D. R. H., Tyrrell, T., Gibbs, S. J., von Dassow, P., Rehm, E., Armbrust,
- 1390 E. V., & Boessenkool, K. P. (2008). Phytoplankton Calcification in a High-CO₂ World. *Science*,
- 1391 320(5874), 336-340. <https://doi.org/10.1126/science.1154122>
- 1392 Iida, Y., Takatani, Y., Kojima, A., & Ishii, M. (2020). Global trends of ocean CO₂ sink and
- 1393 ocean acidification: an observation-based reconstruction of surface ocean inorganic carbon
- 1394 variables. *Journal of Oceanography*, 77(2), 323-358. [https://doi.org/10.1007/s10872-020-](https://doi.org/10.1007/s10872-020-00571-5)
- 1395 00571-5
- 1396 James, A. K., Kelly, L. W., Nelson, C. E., Wilbanks, E. G., & Carlson, C. A. (2019). Elevated
- 1397 pCO₂ alters marine heterotrophic bacterial community composition and metabolic potential in
- 1398 response to a pulse of phytoplankton organic matter. *Environmental Microbiology*, 21(2), 541-
- 1399 556. <https://doi.org/10.1111/1462-2920.14484>
- 1400 Jenrich, M., Angelopoulos, M., Liebner, S., Treat, C., Knoblauch, C., Yang, S., Grosse, G.,
- 1401 Giebler, F., Jongejans, Loeka L., Grigoriev, M., & Strauss, J. (2024). Greenhouse Gas
- 1402 Production and Microbial Response During the Transition From Terrestrial Permafrost to a
- 1403 Marine Environment. *Permafrost and Periglacial Processes*, 36(1), 63-82.
- 1404 <https://doi.org/10.1002/ppp.2251>
- 1405 Jia, F., Sun, D., & Looi, C.-k. (2023). Artificial Intelligence in Science Education (2013–2023):
- 1406 Research Trends in Ten Years. *Journal of Science Education and Technology*, 33(1), 94-117.
- 1407 <https://doi.org/10.1007/s10956-023-10077-6>
- 1408 Jiang, L.-Q., Dunne, J., Carter, B. R., Tjiputra, J. F., Terhaar, J., Sharp, J. D., Olsen, A., Alin,
- 1409 S., Bakker, D. C. E., Feely, R. A., Gattuso, J.-P., Hogan, P., Ilyina, T., Lange, N., Lauvset, S. K.,
- 1410 Lewis, E. R., Lovato, T., Palmieri, J., Santana-Falcón, Y.,...Ziehn, T. (2023). Global Surface
- 1411 Ocean Acidification Indicators From 1750 to 2100. *Journal of Advances in Modeling Earth*
- 1412 *Systems*, 15(3), e2022MS003563. <https://doi.org/https://doi.org/10.1029/2022MS003563>
- 1413 Jin, P., Hutchins, D. A., & Gao, K. (2020). The Impacts of Ocean Acidification on Marine Food

- Quality and Its Potential Food Chain Consequences. *Frontiers in Marine Science*, 7. <https://doi.org/10.3389/fmars.2020.543979>
- Johnson, M. D., Rodriguez Bravo, L. M., O'Connor, S. E., Varley, N. F., & Altieri, A. H. (2019). pH Variability Exacerbates Effects of Ocean Acidification on a Caribbean Crustose Coralline Alga. *Frontiers in Marine Science*, 6. <https://doi.org/10.3389/fmars.2019.00150>
- Karlsson, S., Srebotnjak, T., & Gonzales, P. (2007). Understanding the North–South knowledge divide and its implications for policy: a quantitative analysis of the generation of scientific knowledge in the environmental sciences. *Environmental Science & Policy*, 10(7-8), 668-684. <https://doi.org/10.1016/j.envsci.2007.04.001>
- Kerrison, P., Hall-Spencer, J. M., Suggett, D. J., Hepburn, L. J., & Steinke, M. (2011). Assessment of pH variability at a coastal CO₂ vent for ocean acidification studies. *Estuarine, Coastal and Shelf Science*, 94(2), 129-137. <https://doi.org/10.1016/j.ecss.2011.05.025>
- Kolby Smith, W., Reed, S. C., Cleveland, C. C., Ballantyne, A. P., Anderegg, W. R. L., Wieder, W. R., Liu, Y. Y., & Running, S. W. (2016). Large divergence of satellite and Earth system model estimates of global terrestrial CO₂ fertilization. *Nature Climate Change*, 6(3), 306-310. <https://doi.org/10.1038/nclimate2879>
- Kopf, A., Bicak, M., Kottmann, R., Schnetzer, J., Kostadinov, I., Lehmann, K., Fernandez-Guerra, A., Jeanthon, C., Rahav, E., Ullrich, M., Wichels, A., Gerds, G., Polymenakou, P., Kotoulas, G., Siam, R., Abdallah, R. Z., Sonnenschein, E. C., Cariou, T., O'Gara, F.,... Glöckner, F. O. (2015). The ocean sampling day consortium. *GigaScience*, 4(1), 27. <https://doi.org/10.1186/s13742-015-0066-5>
- Köpsel, V., de Moura Kiipper, G., & Peck, M. A. (2021). Stakeholder engagement vs. social distancing—how does the Covid-19 pandemic affect participatory research in EU marine science projects? *Maritime Studies*, 20(2), 189-205. <https://doi.org/10.1007/s40152-021-00223-4>
- Kroeker, K. J., Kordas, R. L., Crim, R., Hendriks, I. E., Ramajo, L., Singh, G. S., Duarte, C. M., & Gattuso, J.-P. (2013a). Impacts of ocean acidification on marine organisms: quantifying sensitivities and interaction with warming. *Global Change Biology*, 19(6), 1884-1896. <https://doi.org/https://doi.org/10.1111/gcb.12179>
- Kroeker, K. J., Kordas, R. L., Crim, R., Hendriks, I. E., Ramajo, L., Singh, G. S., Duarte, C. M., & Gattuso, J. P. (2013b). Impacts of ocean acidification on marine organisms: quantifying sensitivities and interaction with warming. *Global Change Biology*, 19(6), 1884-1896. <https://doi.org/10.1111/gcb.12179>
- Kroeker, K. J., Kordas, R. L., Crim, R. N., & Singh, G. G. (2010). Meta-analysis reveals negative yet variable effects of ocean acidification on marine organisms. *Ecology Letters*, 13(11), 1419-1434. <https://doi.org/https://doi.org/10.1111/j.1461-0248.2010.01518.x>
- Kroeker, K. J., Kordas, R. L., & Harley, C. D. G. (2017). Embracing interactions in ocean acidification research: confronting multiple stressor scenarios and context dependence. *Biology Letters*, 13(3). <https://doi.org/10.1098/rsbl.2016.0802>
- Kuffner, I. B., Andersson, A. J., Jokiel, P. L., Rodgers, K. u. S., & Mackenzie, F. T. (2007). Decreased abundance of crustose coralline algae due to ocean acidification. *Nature Geoscience*, 1(2), 114-117. <https://doi.org/10.1038/ngeo100>
- Larsen, P. O., & von Ins, M. (2010). The rate of growth in scientific publication and the decline in coverage provided by Science Citation Index. *Scientometrics*, 84(3), 575-603.

- 1458 <https://doi.org/10.1007/s11192-010-0202-z>
- 1459 Leray, M., Wilkins, L. G. E., Apprill, A., Bik, H. M., Clever, F., Connolly, S. R., De León, M.
 1460 E., Duffy, J. E., Ezzat, L., Gignoux-Wolfsohn, S., Herre, E. A., Kaye, J. Z., Kline, D. I.,
 1461 Kueneman, J. G., McCormick, M. K., McMillan, W. O., O'Dea, A., Pereira, T. J., Petersen, J.
 1462 M.,...Eisen, J. A. (2021). Natural experiments and long-term monitoring are critical to
 1463 understand and predict marine host-microbe ecology and evolution. *PLOS Biology*, 19(8).
 1464 <https://doi.org/10.1371/journal.pbio.3001322>
- 1465 Levine, N. M., Alexander, H., Bertrand, E. M., Coles, V. J., Dutkiewicz, S., Leles, S. G., &
 1466 Zakem, E. J. (2025). Microbial Ecology to Ocean Carbon Cycling: From Genomes to
 1467 Numerical Models. *Annual Review of Earth and Planetary Sciences*, 53(Volume 53, 2025),
 1468 595-624. <https://doi.org/https://doi.org/10.1146/annurev-earth-040523-020630>
- 1469 Li, M., & Young, J. N. (2023). Temperature sensitivity of carbon concentrating mechanisms in
 1470 the diatom *Phaeodactylum tricornutum*. *Photosynthesis Research*, 156(2), 205-215.
 1471 <https://doi.org/10.1007/s11120-023-01004-2>
- 1472 Liang, D., Xiang, H., Jin, P., & Xia, J. (2023). Response mechanism of harmful algae
 1473 *Phaeocystis globosa* to ocean warming and acidification. *Environmental Pollution*, 320.
 1474 <https://doi.org/10.1016/j.envpol.2023.121008>
- 1475 Liu, J., Weinbauer, M. G., Maier, C., Dai, M., & Gattuso, J. P. (2010). Effect of ocean
 1476 acidification on microbial diversity and on microbe-driven biogeochemistry and ecosystem
 1477 functioning. *Aquatic Microbial Ecology*, 61(3), 291-305. <https://doi.org/10.3354/ame01446>
- 1478 Liu, W., Ni, R., & Hu, G. (2024). Web of Science Core Collection's coverage expansion: the
 1479 forgotten Arts & Humanities Citation Index? *Scientometrics*, 129(2), 933-955.
 1480 <https://doi.org/10.1007/s11192-023-04917-w>
- 1481 Lohbeck, K. T., Riebesell, U., & Reusch, T. B. H. (2012). Adaptive evolution of a key
 1482 phytoplankton species to ocean acidification. *Nature Geoscience*, 5(5), 346-351.
 1483 <https://doi.org/10.1038/ngeo1441>
- 1484 Lowe, A. T., Bos, J., & Ruesink, J. (2019). Ecosystem metabolism drives pH variability and
 1485 modulates long-term ocean acidification in the Northeast Pacific coastal ocean. *Scientific*
 1486 *Reports*, 9(1). <https://doi.org/10.1038/s41598-018-37764-4>
- 1487 Madsen, E. L. (2011). Microorganisms and their roles in fundamental biogeochemical cycles.
 1488 *Current Opinion in Biotechnology*, 22(3), 456-464.
 1489 <https://doi.org/10.1016/j.copbio.2011.01.008>
- 1490 Madsen, E. L. (2011). Microorganisms and their roles in fundamental biogeochemical cycles.
 1491 *Curr Opin Biotechnol*, 22(3), 456-464. <https://doi.org/10.1016/j.copbio.2011.01.008>
- 1492 Mao, N., Zhu, Y., Wang, J., Shen, X., Hou, J., & Miao, L. (2023). Adaptive Evolution of Marine
 1493 Organisms: A Bibliometric Analysis Based on CiteSpace. *Water*, 15(23).
 1494 <https://doi.org/10.3390/w15234162>
- 1495 Mao, N., Zhu, Y., Wang, J. Y., Shen, X., Hou, J., & Miao, L. Z. (2023). Adaptive Evolution of
 1496 Marine Organisms: A Bibliometric Analysis Based on CiteSpace. *Water*, 15(23), Article 4162.
 1497 <https://doi.org/10.3390/w15234162>
- 1498 Marín-Samper, L., Arístegui, J., Hernández-Hernández, N., Ortiz, J., Archer, S. D., Ludwig, A.,
 1499 & Riebesell, U. (2024). Assessing the impact of CO₂-equilibrated ocean alkalinity
 1500 enhancement on microbial metabolic rates in an oligotrophic system. *Biogeosciences*, 21(11),
 1501 2859-2876. <https://doi.org/10.5194/bg-21-2859-2024>

- 1502 Martins Medeiros, I. P., & Souza, M. M. (2023). Acid times in physiology: A systematic review
1503 of the effects of ocean acidification on calcifying invertebrates. *Environmental Research*, 231,
1504 116019. <https://doi.org/10.1016/j.envres.2023.116019>
- 1505 McCulloch, M. T., D'Olivo, J. P., Falter, J., Holcomb, M., & Trotter, J. A. (2017). Coral
1506 calcification in a changing World and the interactive dynamics of pH and DIC upregulation.
1507 *Nature Communications*, 8(1), 15686. <https://doi.org/10.1038/ncomms15686>
- 1508 McNicholl, C., Koch, M. S., & Hofmann, L. C. (2019). Photosynthesis and light-dependent
1509 proton pumps increase boundary layer pH in tropical macroalgae: A proposed mechanism to
1510 sustain calcification under ocean acidification. *Journal of Experimental Marine Biology and*
1511 *Ecology*, 521. <https://doi.org/10.1016/j.jembe.2019.151208>
- 1512 Meron, D., Rodolfo-Metalpa, R., Cunning, R., Baker, A. C., Fine, M., & Banin, E. (2012).
1513 Changes in coral microbial communities in response to a natural pH gradient. *The ISME Journal*,
1514 6(9), 1775-1785. <https://doi.org/10.1038/ismej.2012.19>
- 1515 Miloslavich, P., Bax, N. J., Simmons, S. E., Klein, E., Appeltans, W., Aburto-Oropeza, O.,
1516 Andersen Garcia, M., Batten, S. D., Benedetti-Cecchi, L., Checkley, D. M., Chiba, S., Duffy, J.
1517 E., Dunn, D. C., Fischer, A., Gunn, J., Kudela, R., Marsac, F., Muller-Karger, F. E., Obura, D.,
1518 & Shin, Y. J. (2018). Essential ocean variables for global sustained observations of biodiversity
1519 and ecosystem changes. *Global Change Biology*, 24(6), 2416-2433.
1520 <https://doi.org/10.1111/gcb.14108>
- 1521 Mongeon, P., & Paul-Hus, A. (2015). The journal coverage of Web of Science and Scopus: a
1522 comparative analysis. *Scientometrics*, 106(1), 213-228. [https://doi.org/10.1007/s11192-015-](https://doi.org/10.1007/s11192-015-1765-5)
1523 1765-5
- 1524 Monteiro, T., Batista, M., & Kerr, R. (2025). Spatial variability of marine carbonate system
1525 along the Drake Passage and northern Antarctic Peninsula during the austral summer. *Marine*
1526 *Chemistry*, 271. <https://doi.org/10.1016/j.marchem.2025.104529>
- 1527 Nawaz, M. Z., Subin Sasidharan, R., Alghamdi, H. A., & Dang, H. (2022). Understanding
1528 Interaction Patterns within Deep-Sea Microbial Communities and Their Potential Applications.
1529 *Marine Drugs*, 20(2). <https://doi.org/10.3390/md20020108>
- 1530 Nelson, K. S., Baltar, F., Lamare, M. D., & Morales, S. E. (2020). Ocean acidification affects
1531 microbial community and invertebrate settlement on biofilms. *Scientific Reports*, 10(1).
1532 <https://doi.org/10.1038/s41598-020-60023-4>
- 1533 Newbold, L. K., Oliver, A. E., Booth, T., Tiwari, B., DeSantis, T., Maguire, M., Andersen, G.,
1534 van der Gast, C. J., & Whiteley, A. S. (2012). The response of marine picoplankton to ocean
1535 acidification. *Environmental Microbiology*, 14(9), 2293-2307. [https://doi.org/10.1111/j.1462-](https://doi.org/10.1111/j.1462-2920.2012.02762.x)
1536 2920.2012.02762.x
- 1537 O'Brien, P. A., Morrow, K. M., Willis, B. L., & Bourne, D. G. (2016). Implications of Ocean
1538 Acidification for Marine Microorganisms from the Free-Living to the Host-Associated.
1539 *Frontiers in Marine Science*, 3. <https://doi.org/10.3389/fmars.2016.00047>
- 1540 Olson, E. M., John, J. G., Dunne, J. P., Stock, C. A., & Drenkard, E. J. (2025). Potential for
1541 Regional Resilience to Ocean Warming and Acidification Extremes: Projected Vulnerability
1542 Under Contrasting Pathways and Thresholds. *Global Change Biology*, 31(7).
1543 <https://doi.org/10.1111/gcb.70360>
- 1544 Orr, J. C., Fabry, V. J., Aumont, O., Bopp, L., Doney, S. C., Feely, R. A., Gnanadesikan, A.,
1545 Gruber, N., Ishida, A., Joos, F., Key, R. M., Lindsay, K., Maier-Reimer, E., Matear, R., Monfray,

- 1546 P., Mouchet, A., Najjar, R. G., Plattner, G.-K., Rodgers, K. B.,... Yool, A. (2005). Anthropogenic
 1547 ocean acidification over the twenty-first century and its impact on calcifying organisms. *Nature*,
 1548 437(7059), 681-686. <https://doi.org/10.1038/nature04095>
- 1549 Osborne, E., Hu, X., Hall, E. R., Yates, K., Vreeland-Dawson, J., Shamberger, K., Barbero, L.,
 1550 Martin Hernandez-Ayon, J., Gomez, F. A., Hicks, T., Xu, Y.-Y., McCutcheon, M. R.,
 1551 Acquafredda, M., Chapa-Balcorta, C., Norzagaray, O., Pierrot, D., Munoz-Caravaca, A.,
 1552 Dobson, K. L., Williams, N.,... Dash, P. (2022). Ocean acidification in the Gulf of Mexico:
 1553 Drivers, impacts, and unknowns. *Progress in Oceanography*, 209.
 1554 <https://doi.org/10.1016/j.pocean.2022.102882>
- 1555 Pasgaard, M., Dalsgaard, B., Maruyama, P. K., Sandel, B., & Strange, N. (2015). Geographical
 1556 imbalances and divides in the scientific production of climate change knowledge. *Global*
 1557 *Environmental Change*, 35, 279-288. <https://doi.org/10.1016/j.gloenvcha.2015.09.018>
- 1558 Piontek, J., Lunau, M., Händel, N., Borchard, C., Wurst, M., & Engel, A. (2010). Acidification
 1559 increases microbial polysaccharide degradation in the ocean. *Biogeosciences*, 7(5), 1615-1624.
 1560 <https://doi.org/10.5194/bg-7-1615-2010>
- 1561 Prancutè, R. (2021). Web of Science (WoS) and Scopus: The Titans of Bibliographic
 1562 Information in Today's Academic World. *Publications*, 9(1).
 1563 <https://doi.org/10.3390/publications9010012>
- 1564 Qiu, L., Esposito, M., Martínez-Cabanas, M., Achterberg, E. P., & Li, Q. (2023). Autonomous
 1565 high-frequency time-series observations of total alkalinity in dynamic estuarine waters. *Marine*
 1566 *Chemistry*, 257. <https://doi.org/10.1016/j.marchem.2023.104332>
- 1567 Rastelli, E., Petani, B., Corinaldesi, C., Dell'Anno, A., Lo Martire, M., Cerrano, C., &
 1568 Danovaro, R. (2020). A high biodiversity mitigates the impact of ocean acidification on hard-
 1569 bottom ecosystems. *Scientific Reports*, 10(1), 2948. [https://doi.org/10.1038/s41598-020-](https://doi.org/10.1038/s41598-020-59886-4)
 1570 59886-4
- 1571 Raven, J. A., Beardall, J., & Giordano, M. (2014). Energy costs of carbon dioxide concentrating
 1572 mechanisms in aquatic organisms. *Photosynthesis Research*, 121(2), 111-124.
 1573 <https://doi.org/10.1007/s11120-013-9962-7>
- 1574 Raven, J. A., & Crawford, K. (2012). Environmental controls on coccolithophore calcification.
 1575 *Marine Ecology Progress Series*, 470, 137-166. <https://doi.org/10.3354/meps09993>
- 1576 Raven, J. A., Giordano, M., Beardall, J., & Maberly, S. C. (2011). Algal and aquatic plant carbon
 1577 concentrating mechanisms in relation to environmental change. *Photosynthesis Research*,
 1578 109(1-3), 281-296. <https://doi.org/10.1007/s11120-011-9632-6>
- 1579 Rees, A. P., Bange, H. W., Arévalo-Martínez, D. L., Artioli, Y., Ashby, D. M., Brown, I.,
 1580 Campen, H. I., Clark, D. R., Kitidis, V., Lessin, G., Tarran, G. A., & Turley, C. (2021). Nitrous
 1581 oxide and methane in a changing Arctic Ocean. *Ambio*, 51(2), 398-410.
 1582 <https://doi.org/10.1007/s13280-021-01633-8>
- 1583 Reinfelder, J. R. (2011). Carbon concentrating mechanisms in eukaryotic marine phytoplankton.
 1584 *Ann Rev Mar Sci*, 3, 291-315. <https://doi.org/10.1146/annurev-marine-120709-142720>
- 1585 Révelard, A., Tintoré, J., Verron, J., Baharel, P., Barth, J. A., Belbéoch, M., Benveniste, J.,
 1586 Bonnefond, P., Chassignet, E. P., Cravatte, S., Davidson, F., deYoung, B., Heupel, M., Heslop,
 1587 E., Hörstmann, C., Karstensen, J., Le Traon, P. Y., Marques, M., McLean, C.,... Williams, B.
 1588 (2022). Ocean Integration: The Needs and Challenges of Effective Coordination Within the
 1589 Ocean Observing System. *Frontiers in Marine Science*, 8.

- 1590 <https://doi.org/10.3389/fmars.2021.737671>
- 1591 Ridgwell, A., Schmidt, D. N., Turley, C., Brownlee, C., Maldonado, M. T., Tortell, P., & Young,
1592 J. R. (2009). From laboratory manipulations to Earth system models: scaling calcification
1593 impacts of ocean acidification. *Biogeosciences*, 6(11), 2611-2623. [https://doi.org/10.5194/bg-](https://doi.org/10.5194/bg-6-2611-2009)
1594 6-2611-2009
- 1595 Riebesell, U., Czerny, J., von Bröckel, K., Boxhammer, T., Büdenbender, J., Deckelnick, M.,
1596 Fischer, M., Hoffmann, D., Krug, S. A., Lentz, U., Ludwig, A., Mücke, R., & Schulz, K. G.
1597 (2013). Technical Note: A mobile sea-going mesocosm system – new opportunities for ocean
1598 change research. *Biogeosciences*, 10(3), 1835-1847. <https://doi.org/10.5194/bg-10-1835-2013>
- 1599 Riebesell, U., & Gattuso, J.-P. (2014). Lessons learned from ocean acidification research.
1600 *Nature Climate Change*, 5(1), 12-14. <https://doi.org/10.1038/nclimate2456>
- 1601 Riebesell, U., Schulz, K. G., Bellerby, R. G. J., Botros, M., Fritsche, P., Meyerhöfer, M., Neill,
1602 C., Nondal, G., Oschlies, A., Wohlers, J., & Zöllner, E. (2007). Enhanced biological carbon
1603 consumption in a high CO₂ ocean. *Nature*, 450(7169), 545-548.
1604 <https://doi.org/10.1038/nature06267>
- 1605 Riebesell, U., Zondervan, I., Rost, B., Tortell, P. D., Zeebe, R. E., & Morel, F. M. M. (2000).
1606 Reduced calcification of marine plankton in response to increased atmospheric CO₂. *Nature*,
1607 407(6802), 364-367. <https://doi.org/10.1038/35030078>
- 1608 Robinson, A. R., McCarthy, J. J., & Rothschild, B. J. (1999). Interdisciplinary ocean science is
1609 evolving and a systems approach is essential. *Journal of Marine Systems*, 22(4), 231-239.
1610 [https://doi.org/10.1016/s0924-7963\(99\)00064-0](https://doi.org/10.1016/s0924-7963(99)00064-0)
- 1611 Rost, B., Zondervan, I., & Wolf-Gladrow, D. (2008). Sensitivity of phytoplankton to future
1612 changes in ocean carbonate chemistry: current knowledge, contradictions and research
1613 directions. *Marine Ecology Progress Series*, 373, 227-237. <https://doi.org/10.3354/meps07776>
- 1614 Rubbens, P., Brodie, S., Cordier, T., Destro Barcellos, D., Devos, P., Fernandes-Salvador, J. A.,
1615 Fincham, J. I., Gomes, A., Handegard, N. O., Howell, K., Jamet, C., Kartveit, K. H., Moustahfid,
1616 H., Parcerisas, C., Politikos, D., Sauzède, R., Sokolova, M., Uusitalo, L., Van den Bulcke,
1617 L.,... Whidden, C. (2023). Machine learning in marine ecology: an overview of techniques and
1618 applications. *Ices Journal of Marine Science*, 80(7), 1829-1853.
1619 <https://doi.org/10.1093/icesjms/fsad100>
- 1620 Salomon, A. K., & McKechnie, I. (2025). Insights Gained from Including People in Our Models
1621 of Nature and Modes of Science. *Annual Review of Marine Science*, 17(1), 167-191.
1622 <https://doi.org/10.1146/annurev-marine-021523-105524>
- 1623 Sastri, A. R., Christian, J. R., Achterberg, E. P., Atamanchuk, D., Buck, J. J. H., Bresnahan, P.,
1624 Duke, P. J., Evans, W., Gonski, S. F., Johnson, B., Juniper, S. K., Mihaly, S., Miller, L. A.,
1625 Morley, M., Murphy, D., Nakaoka, S.-i., Ono, T., Parker, G., Simpson, K., & Tsunoda, T. (2019).
1626 Perspectives on in situ Sensors for Ocean Acidification Research. *Frontiers in Marine Science*,
1627 6. <https://doi.org/10.3389/fmars.2019.00653>
- 1628 Schönberg, C. H. L., Fang, J. K. H., Carreiro-Silva, M., Tribollet, A., & Wisshak, M. (2017).
1629 Bioerosion: the other ocean acidification problem. *ICES Journal of Marine Science*, 74(4), 895-
1630 925. <https://doi.org/10.1093/icesjms/fsw254>
- 1631 Schulz, K. G., Bellerby, R. G. J., Brussaard, C. P. D., Büdenbender, J., Czerny, J., Engel, A.,
1632 Fischer, M., Koch-Klavsén, S., Krug, S. A., Lischka, S., Ludwig, A., Meyerhöfer, M., Nondal,
1633 G., Silyakova, A., Stühr, A., & Riebesell, U. (2013). Temporal biomass dynamics of an Arctic

- 1634 plankton bloom in response to increasing levels of atmospheric carbon dioxide. *Biogeosciences*,
 1635 10(1), 161-180. <https://doi.org/10.5194/bg-10-161-2013>
- 1636 Simone, M. N., Schulz, K. G., Oakes, J. M., & Eyre, B. D. (2021). Warming and ocean
 1637 acidification may decrease estuarine dissolved organic carbon export to the ocean.
 1638 *Biogeosciences*, 18(5), 1823-1838. <https://doi.org/10.5194/bg-18-1823-2021>
- 1639 Simonetti, S., Zupo, V., Gambi, M. C., Luckenbach, T., & Corsi, I. (2022). Unraveling cellular
 1640 and molecular mechanisms of acid stress tolerance and resistance in marine species: New
 1641 frontiers in the study of adaptation to ocean acidification. *Marine Pollution Bulletin*, 185.
 1642 <https://doi.org/10.1016/j.marpolbul.2022.114365>
- 1643 Spilling, K., Schulz, K. G., Paul, A. J., Boxhammer, T., Achterberg, E. P., Hornick, T., Lischka,
 1644 S., Stühr, A., Bermúdez, R., Czerny, J., Crawford, K., Brussaard, C. P. D., Grossart, H.-P., &
 1645 Riebesell, U. (2016). Effects of ocean acidification on pelagic carbon fluxes in a mesocosm
 1646 experiment. *Biogeosciences*, 13(21), 6081-6093. <https://doi.org/10.5194/bg-13-6081-2016>
- 1647 Spisla, C., Taucher, J., Bach, L. T., Haunost, M., Boxhammer, T., King, A. L., Jenkins, B. D.,
 1648 Wallace, J. R., Ludwig, A., Meyer, J., Stange, P., Minutolo, F., Lohbeck, K. T., Nauendorf, A.,
 1649 Kalter, V., Lischka, S., Sswat, M., Dörner, I., Ismar-Rebitz, S. M. H.,...Riebesell, U. (2021).
 1650 Extreme Levels of Ocean Acidification Restructure the Plankton Community and
 1651 Biogeochemistry of a Temperate Coastal Ecosystem: A Mesocosm Study. *Frontiers in Marine*
 1652 *Science*, 7. <https://doi.org/10.3389/fmars.2020.611157>
- 1653 Steffen, W., Leinfelder, R., Zalasiewicz, J., Waters, C. N., Williams, M., Summerhayes, C.,
 1654 Barnosky, A. D., Cearreta, A., Crutzen, P., Edgeworth, M., Ellis, E. C., Fairchild, I. J., Galuszka,
 1655 A., Grinevald, J., Haywood, A., Ivar do Sul, J., Jeandel, C., McNeill, J. R., Odada,
 1656 E.,...Schellnhuber, H. J. (2016). Stratigraphic and Earth System approaches to defining the
 1657 Anthropocene. *Earth's Future*, 4(8), 324-345. <https://doi.org/10.1002/2016ef000379>
- 1658 Stockdale, A., Tipping, E., Loftis, S., & Mortimer, R. J. G. (2016). Effect of Ocean Acidification
 1659 on Organic and Inorganic Speciation of Trace Metals. *Environmental Science & Technology*,
 1660 50(4), 1906-1913. <https://doi.org/10.1021/acs.est.5b05624>
- 1661 Subramaniam, R. C., Melbourne-Thomas, J., Davidson, A. T., & Corney, S. P. (2016).
 1662 Mechanisms driving Antarctic microbial community responses to ocean acidification: a
 1663 network modelling approach. *Polar Biology*, 40(3), 727-734. [https://doi.org/10.1007/s00300-](https://doi.org/10.1007/s00300-016-1989-8)
 1664 016-1989-8
- 1665 Sutton, A. J., Battisti, R., Carter, B., Evans, W., Newton, J., Alin, S., Bates, N. R., Cai, W.-J.,
 1666 Currie, K., Feely, R. A., Sabine, C., Tanhua, T., Tilbrook, B., & Wanninkhof, R. (2022).
 1667 Advancing best practices for assessing trends of ocean acidification time series. *Frontiers in*
 1668 *Marine Science*, 9. <https://doi.org/10.3389/fmars.2022.1045667>
- 1669 Tagliabue, A., Bopp, L., & Gehlen, M. (2011). The response of marine carbon and nutrient
 1670 cycles to ocean acidification: Large uncertainties related to phytoplankton physiological
 1671 assumptions. *Global Biogeochemical Cycles*, 25(3), n/a-n/a.
 1672 <https://doi.org/10.1029/2010gb003929>
- 1673 Tangherlini, M., Corinaldesi, C., Ape, F., Greco, S., Romeo, T., Andaloro, F., & Danovaro, R.
 1674 (2021). Ocean Acidification Induces Changes in Virus-Host Relationships in Mediterranean
 1675 Benthic Ecosystems. *Microorganisms*, 9(4), Article 769.
 1676 <https://doi.org/10.3390/microorganisms9040769>
- 1677 Taucher, J., Boxhammer, T., Bach, L. T., Paul, A. J., Schartau, M., Stange, P., & Riebesell, U.

- (2020). Changing carbon-to-nitrogen ratios of organic-matter export under ocean acidification. *Nature Climate Change*, 11(1), 52-57. <https://doi.org/10.1038/s41558-020-00915-5>
- Tilbrook, B., Jewett, E. B., DeGrandpre, M. D., Hernandez-Ayon, J. M., Feely, R. A., Gledhill, D. K., Hansson, L., Isensee, K., Kurz, M. L., Newton, J. A., Siedlecki, S. A., Chai, F., Dupont, S., Graco, M., Calvo, E., Greeley, D., Kapsenberg, L., Lebrech, M., Pelejero, C.,...Telszewski, M. (2019). An Enhanced Ocean Acidification Observing Network: From People to Technology to Data Synthesis and Information Exchange [Policy and Practice Reviews]. *Frontiers in Marine Science*, Volume 6 - 2019. <https://doi.org/10.3389/fmars.2019.00337>
- Treusch, A. H., Vergin, K. L., Finlay, L. A., Donatz, M. G., Burton, R. M., Carlson, C. A., & Giovannoni, S. J. (2009). Seasonality and vertical structure of microbial communities in an ocean gyre. *The ISME Journal*, 3(10), 1148-1163. <https://doi.org/10.1038/ismej.2009.60>
- Trull, T. W., Passmore, A., Davies, D. M., Smit, T., Berry, K., & Tilbrook, B. (2018). Distribution of planktonic biogenic carbonate organisms in the Southern Ocean south of Australia: a baseline for ocean acidification impact assessment. *Biogeosciences*, 15(1), 31-49. <https://doi.org/10.5194/bg-15-31-2018>
- Uchikawa, J., & Zeebe, R. E. (2008). Influence of terrestrial weathering on ocean acidification and the next glacial inception. *Geophysical Research Letters*, 35(23). <https://doi.org/10.1029/2008gl035963>
- Ugolini, G. S., Wang, M., Secchi, E., Pioli, R., Ackermann, M., & Stocker, R. (2024). Microfluidic approaches in microbial ecology [10.1039/D3LC00784G]. *Lab on a Chip*, 24(5), 1394-1418. <https://doi.org/10.1039/D3LC00784G>
- Velthuis, M., Keuskamp, J. A., Bakker, E. S., Boersma, M., Sommer, U., van Donk, E., & Van de Waal, D. B. (2022). Differential effects of elevated pCO₂ and warming on marine phytoplankton stoichiometry. *Limnology and Oceanography*, 67(3), 598-607. <https://doi.org/10.1002/lno.12020>
- Vodopivec, M. (2024). A numerical model sheds new light on the effects of ocean acidification on an indicator species. *Global Change Biology*, 30(7). <https://doi.org/10.1111/gcb.17398>
- Wang, B., Hua, L., Mei, H., Wu, X., Kang, Y., & Zhao, N. (2024). Impact of Climate Change on the Dynamic Processes of Marine Environment and Feedback Mechanisms: An Overview. *Archives of Computational Methods in Engineering*, 31(6), 3377-3408. <https://doi.org/10.1007/s11831-024-10072-z>
- Wang, T., Jin, P., Wells, M. L., Trick, C. G., & Gao, K. (2019). Insensitivities of a subtropical productive coastal plankton community and trophic transfer to ocean acidification: Results from a microcosm study. *Marine Pollution Bulletin*, 141, 462-471. <https://doi.org/10.1016/j.marpolbul.2019.03.002>
- Wang, Y., Zhang, R., Yang, Y., Tu, Q., Zhou, J., & Jiao, N. (2023). Ocean acidification altered microbial functional potential in the Arctic Ocean. *Limnology and Oceanography*, 68(S1). <https://doi.org/10.1002/lno.12375>
- Wang, Z., Tsementzi, D., Williams, T. C., Juarez, D. L., Blinbry, S. K., Garcia, N. S., Sienkiewicz, B. K., Konstantinidis, K. T., Johnson, Z. I., & Hunt, D. E. (2021). Environmental stability impacts the differential sensitivity of marine microbiomes to increases in temperature and acidity. *The ISME Journal*, 15(1), 19-28. <https://doi.org/10.1038/s41396-020-00748-2>
- Wannicke, N., Frey, C., Law, C. S., & Voss, M. (2018). The response of the marine nitrogen cycle to ocean acidification. *Global Change Biology*, 24(11), 5031-5043.

- 1722 <https://doi.org/10.1111/gcb.14424>
- 1723 Webster, N. S., Negri, A. P., Botté, E. S., Laffy, P. W., Flores, F., Noonan, S., Schmidt, C., &
 1724 Uthicke, S. (2016). Host-associated coral reef microbes respond to the cumulative pressures of
 1725 ocean warming and ocean acidification. *Scientific Reports*, 6(1).
 1726 <https://doi.org/10.1038/srep19324>
- 1727 Wei, J., Liang, G., Alex, J., Zhang, T., & Ma, C. (2020). Research Progress of Energy Utilization
 1728 of Agricultural Waste in China: Bibliometric Analysis by Citespace. *Sustainability*, 12(3).
 1729 <https://doi.org/10.3390/su12030812>
- 1730 Whitehead, A. (2012). Comparative genomics in ecological physiology: toward a more nuanced
 1731 understanding of acclimation and adaptation. *Journal of Experimental Biology*, 215(6), 884-
 1732 891. <https://doi.org/10.1242/jeb.058735>
- 1733 Witt, V., Wild, C., Anthony, K. R. N., Diaz-Pulido, G., & Uthicke, S. (2011). Effects of ocean
 1734 acidification on microbial community composition of, and oxygen fluxes through, biofilms
 1735 from the Great Barrier Reef. *Environmental Microbiology*, 13(11), 2976-2989.
 1736 <https://doi.org/10.1111/j.1462-2920.2011.02571.x>
- 1737 Wittmann, A. C., & Pörtner, H.-O. (2013). Sensitivities of extant animal taxa to ocean
 1738 acidification. *Nature Climate Change*, 3(11), 995-1001. <https://doi.org/10.1038/nclimate1982>
- 1739 Wu, H. C., Dissard, D., Douville, E., Blamart, D., Bordier, L., Tribollet, A., Le Cornec, F., Pons-
 1740 Branchu, E., Dapoigny, A., & Lazareth, C. E. (2018). Surface ocean pH variations since 1689
 1741 CE and recent ocean acidification in the tropical South Pacific. *Nature Communications*, 9(1).
 1742 <https://doi.org/10.1038/s41467-018-04922-1>
- 1743 Xu, J., Sun, J., Beardall, J., & Gao, K. (2020). Lower Salinity Leads to Improved Physiological
 1744 Performance in the Coccolithophorid *Emiliania huxleyi*, Which Partly Ameliorates the Effects
 1745 of Ocean Acidification [Original Research]. *Frontiers in Marine Science*, Volume 7 - 2020.
 1746 <https://doi.org/10.3389/fmars.2020.00704>
- 1747 Young, J. N., Kranz, S. A., Goldman, J. A. L., Tortell, P. D., & Morel, F. M. M. (2015). Antarctic
 1748 phytoplankton down-regulate their carbon-concentrating mechanisms under high CO₂ with no
 1749 change in growth rates. *Marine Ecology Progress Series*, 532, 13-28. [https://www.int-
 1750 res.com/abstracts/meps/v532/meps11336](https://www.int-res.com/abstracts/meps/v532/meps11336)
- 1751 Zark, M., Riebesell, U., & Dittmar, T. (2015a). Effects of ocean acidification on marine
 1752 dissolved organic matter are not detectable over the succession of phytoplankton blooms.
 1753 *Science Advances*, 1(9). <https://doi.org/10.1126/sciadv.1500531>
- 1754 Zark, M., Riebesell, U., & Dittmar, T. (2015b). Effects of ocean acidification on marine
 1755 dissolved organic matter are not detectable over the succession of phytoplankton blooms.
 1756 *Science Advances*, 1(9), e1500531. <https://doi.org/doi:10.1126/sciadv.1500531>
- 1757 Zeebe, R. E. (2012). History of Seawater Carbonate Chemistry, Atmospheric CO₂, and Ocean
 1758 Acidification. *Annual Review of Earth and Planetary Sciences*, 40(1), 141-165.
 1759 <https://doi.org/10.1146/annurev-earth-042711-105521>
- 1760 Zhang, X., Chen, S., Yao, P., Han, J., & Jin, R. (2025). Knowledge Structure and Evolution of
 1761 Wetland Plant Diversity Research: Visual Exploration Based on CiteSpace. *Biology*, 14(7).
 1762 <https://doi.org/10.3390/biology14070781>
- 1763 Zhang, X., Xi, T., Wang, Y., Fan, X., Xu, D., Zhang, P., Sun, K., Zhang, Y., Ma, J., & Ye, N.
 1764 (2024). Chemical interactions between kelp *Macrocystis pyrifera* and symbiotic bacteria under
 1765 elevated CO₂ condition. *Marine Life Science & Technology*, 6(4), 700-712.

- 1766 <https://doi.org/10.1007/s42995-024-00259-5>
- 1767 Zhang, Y., Beardall, J., & Gao, K. (2025). Combined effects of ocean acidification and warming
- 1768 on phytoplankton productivity and community structure in the coastal water of Southern East.
- 1769 *Marine Environmental Research*, 210, 107352.
- 1770 <https://doi.org/https://doi.org/10.1016/j.marenvres.2025.107352>
- 1771 Zhou, J., Zheng, Y., Hou, L., Qi, L., Mao, T., Yin, G., & Liu, M. (2024). Nitrogen input
- 1772 modulates the effects of coastal acidification on nitrification and associated N₂O emission.
- 1773 *Water Research*, 261, 122041. <https://doi.org/https://doi.org/10.1016/j.watres.2024.122041>
- 1774 Zunino, S., Libralato, S., Melaku Canu, D., Prato, G., & Solidoro, C. (2021). Impact of Ocean
- 1775 Acidification on Ecosystem Functioning and Services in Habitat-Forming Species and Marine
- 1776 Ecosystems. *Ecosystems*, 24(7), 1561-1575. <https://doi.org/10.1007/s10021-021-00601-3>

1777

Declaration of interests

☒ The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

☐ The authors declare the following financial interests/personal relationships which may be considered as potential competing interests: